

BIOLOGY GRADE 11: GROWTH AND DEVELOPMENT IN PLANTS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.2 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	11
Subject	Biology
Strand	Strand 2.0: Anatomy and Physiology of Plants
Sub-Strand	Sub-Strand 2.2: Growth and Development in Plants
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Meaning and significance of growth and development – Structure of the seed (monocots and dicots) – Seed dormancy (meaning, significance, causes and ways of breaking seed dormancy) – Germination (process, types and conditions necessary for germination) – Primary and secondary growth – Measurement of growth and growth curves – Role of hormones in plant growth and development (include apical dominance)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - distinguish between growth and development in living organisms, - relate structures of monocot and dicot seeds to their functions, - outline causes and ways of breaking dormancy in seeds, - investigate conditions necessary for seed germination, - investigate types of germination in plants, - illustrate primary and secondary growth in flowering plants, - demonstrate the role of hormones in plant growth and development, - appreciate the importance of growth and development in plants.
Core Competencies	– Citizenship: The learner solves problems in the community by sharing the seedlings from the established tree nursery. – Learning to Learn: The learner reflects on their own work and adjusts accordingly while carrying out activities to demonstrate ways of breaking seed dormancy. – Critical Thinking and Problem Solving: Learners investigate the farmer's problem of uneven germination and growth of maize and beans. – Communication and Collaboration: Learners share evidence and findings in group investigations. – Creativity and Imagination: Learners design experiments and build and revise models of germination and growth. – Digital Literacy: Learners use the maize and bean time-lapse videos and other digital resources, with teacher-prepared alternatives when offline.
Core Values	– Unity: The learner works harmoniously with others while carrying out experiments to investigate conditions necessary for seed

	germination. – Patriotism: The learner demonstrates devotion to the country/community while sharing seedlings with the school community for environmental conservation, and links healthy plant growth to Kenya's food security. – Responsibility: Learners care for the experimental seeds and follow procedures. – Integrity: Learners record actual observations rather than changing results to match predictions. – Respect: Learners listen to and consider other learners' explanations.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners post questions about the maize and bean phenomenon on the Driving Question Board. – Developing and Using Models: Learners draw an initial model in Lesson 1 and revise it as they learn about seeds, dormancy, germination, growth and hormones. – Planning and Carrying Out Investigations: Learners test conditions for germination, compare hypogeal and epigeal germination, and break dormancy using maize and bean seeds. – Analyzing and Interpreting Data: Learners record seedling height and stem width and draw growth curves. – Constructing Explanations: Learners explain the differences in germination, emergence and growth using evidence. – Engaging in Argument from Evidence: Learners defend explanations and respond to peers' ideas. – Obtaining, Evaluating and Communicating Information: Learners use readings, videos and microscope sections to share findings.
Pertinent & Contemporary Issues (PCIs)	– Financial Literacy: The learner explores ways of generating income while selling seedlings to members of the community. – Prevention of Global Warming: The learner mitigates the effects of global warming by establishing a tree nursery and distributing seedlings in the community. – Food security and nutrition, sustainable agriculture and agricultural productivity: Understanding germination and growth helps improve crop establishment and yields of maize and beans. – Climate change and adaptation, environmental and water conservation: Learners relate rainfall and soil moisture to germination and growth. – Technology and innovation: Learners use time-lapse videos and plant hormone applications in agriculture.
Career Connections	– Agronomist: How can farmers improve crop establishment and growth? – Seed Technologist: Why do some seeds germinate while others fail to germinate? – Agricultural Research Scientist: What conditions produce better germination and plant growth? – Plant Physiologist: What processes and hormones control plant growth? – Plant Breeder: Can varieties be developed that germinate and grow better under Kenyan conditions? – Agricultural Extension Officer: How can this knowledge be transferred to farmers?
Focus for Lessons	This unit uses a farmer's maize and bean field after rainfall as an anchoring phenomenon to explain why seeds germinate at different times, some fail to germinate, and plants grow differently. Learners investigate seed structure, dormancy, germination, primary and secondary growth, growth measurement and hormones using locally available seeds and low-cost materials. They revise a cumulative model and use the Driving Question Board throughout, and the unit ends with a tree nursery project using natural rooting hormones.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: What causes the differences in germination, emergence and growth of maize and bean plants? KICD KEY INQUIRY QUESTIONS: 1. How does growth occur in plants?

Anchoring Phenomenon	A farmer plants maize and beans on the same day after rainfall. Time-lapse videos show the seeds germinating and the seedlings growing. Some seeds germinate within a few days, while others take longer, and some fail to germinate even though they appear healthy. As the plants grow, stems become taller and thicker, roots extend deeper, and some plants develop branches. It is puzzling because the seeds were planted together under the same rain, yet they behave differently. Learners must explain what causes these differences in germination, emergence and growth.
Storyline Thread	Lesson 1: The learner gets to know the phenomena. Watch the time-lapse videos of maize and beans (Bean Time-Lapse - 25 days Soil cross section; Growing corn time lapse), draw initial models, write questions and post them on the Driving Question Board. Question raised: What is growth and development? Lesson 2: Meaning of growth and development. Structure of the seed (monocot and dicot). Discuss the meaning of growth and development, then observe, draw and label monocot and dicot seeds and discuss the functions of the structures with peers. Question raised: What is seed dormancy? Lesson 3: Seed dormancy. Examine seed specimens to identify causes of seed dormancy and discuss with peers; carry out activities to demonstrate ways of breaking seed dormancy (Khan Academy video on seed dormancy where available). Question raised: How does germination occur? Lesson 4: Germination (process, types and conditions necessary for germination). Use maize and bean seeds to investigate conditions necessary for germination and carry out experiments to investigate types of germination. Question raised: How do plants show growth? Lesson 5: Germination continued. Set up and analyze the germination investigations, compare hypogeal and epigeal germination, and revise models. Question raised: How do plants show growth? Lesson 6: Primary and secondary growth. Observe sections of stems under a microscope or photomicrographs, draw and label them, and use drawings and models to illustrate stages of secondary growth. Question raised: How can you measure growth in plants? Lesson 7: Primary and secondary growth continued. Model secondary growth in a woody dicot, and use seedlings to determine regions of growth in plants and discuss with peers. Question raised: How can you measure growth in plants? Lesson 8: Measurement of growth and growth curves. Record seedling height and stem width data, draw growth curves and discuss with peers. Question raised: Do we have hormones in plants? Lesson 9: Role of hormones in plant growth and development. Watch a video showing the effects of plant hormones, read on the functions of major plant hormones including apical dominance, and relate them to agriculture. Lesson 10: Synthesis and application. Complete the final cumulative model, revisit the Driving Question Board, explain the farmer's maize and bean differences, and plan the tree nursery project using stem cuttings and natural rooting hormones such as bulb onion, Aloe and cinnamon.
Supporting Phenomena	– Maize seedlings (monocot) emerge with the seed remaining underground, while bean seedlings (dicot) push up with seed leaves above the soil. – Healthy-looking seeds fail to germinate until the seed coat is scratched or soaked in water, as with some hard-coated seeds. – A tree trunk becomes thicker each year while the seedling tip keeps growing taller, shown by annual rings and elongating shoots. – Pinching off the tip of a plant such as a bean or tea shoot makes side branches grow, while a stem cutting treated with bulb onion or Aloe extract grows roots.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes (double period) or 2 x 40 minutes): The Farmer's Puzzle: Why Do Maize and Beans Behave Differently After the Same Rain?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon: maize and beans planted on the same day after rainfall germinate, emerge and grow differently. Learners watch the time-lapse videos, draw initial models, write baseline explanations and generate questions that open the Driving Question Board (DQB).
Knowledge	Learners begin to recognise that seeds planted together under the same rain can differ in germination time, emergence and growth pattern (stem height and thickness, root depth, branching). They name what they notice and wonder about, and identify the question to pursue next: What is growth and development? This links to the sub-strand outcome: distinguish between growth and development in living organisms.
Skills	Developing and using models (drawing and labelling initial models of a maize seed and a bean seed); Asking questions; Observing and recording patterns from time-lapse video; Constructing explanations (initial, unmarked); Communicating ideas in simple, clear scientific English.
Attitudes	Curiosity and willingness to wonder; respect for others' ideas (Unity: the learner works harmoniously with others); honesty in sharing baseline thinking without fear of being wrong; appreciation of the importance of growth and development in plants. Learning to Learn: the learner reflects on their own ideas and expects to adjust them later.
Key Inquiry Question	How does growth occur in plants?
Purpose in Storyline	Lesson 1 introduces the phenomenon and creates the need to know. Learners' baseline models and questions go onto the DQB and drive every later lesson: seed structure, dormancy, conditions for germination, types of germination, primary and secondary growth, growth curves and hormones. Learners return to these Day 1 models at the end of the unit to revise them with evidence. The lesson ends with the new question: What is growth and development?
Safety Notes	If learners handle real seeds, they should wash hands afterwards and must not eat seeds that are treated (dressed) with pesticide, often pink or red. Keep seeds away from the mouth. Arrange seating so learners can see the screen or demonstration without crowding. Keep walkways clear during DQB posting.

B. LESSON OVERVIEW

Teacher opens with a short story about a farmer in Nakuru County (Rift Valley) who plants maize and beans on the same day after the long rains and is puzzled by the results. Learners predict what will happen, then watch two time-lapse videos (Bean Time-Lapse - 25 days | Soil cross section; Growing corn time lapse #greentimelapse #gtl #timelapse). If there is no electricity or internet, the teacher uses a prepared alternative: a sequence of 6 to 8 large chalk or chart drawings or photographs showing day 0, 3, 7, 14 and 25, plus a teacher-narrated script, and real soaked and sprouted maize and bean seeds on a tray. Learners record 'I notice' and 'I wonder' statements using Think-Pair-Share, then individually draw initial models of a maize seed and a bean seed, showing what happens after planting, the root and shoot emerging, and development over time, with arrows and labels. They complete the sentence starters, which are not marked right or wrong. In small groups they write questions on sticky notes or cards and post them on the DQB, sorting them into categories. The class agrees on the first new question, 'What is growth and development?', which leads into Lesson 2. Learners keep their drawings in a folder for revision at the end of the unit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Exponential growth & decay word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  H5P: Tra's Red Bean Plant Source: Global Digital Library - Book Catalog  Search ARES for similar readings	Learners listen to the farmer's story: 'Mama Wanjiru in Molo plants maize and beans on the same morning after the first heavy rains. The seeds look healthy. After one week, some seeds have sprouted, some are late, and a few have not come up at all.' Learners individually write a quick prediction in their notebooks: 'I think the maize and the beans will _____ because _____. ' They also complete 'I think a seed needs _____ in order to germinate because...'. Then they Think-Pair-Share with a partner, comparing which seeds they think will come up first and why. They share predictions with the class.	Read the story slowly with a local accent and actions, then say: 'Mama Wanjiru planted maize and beans on the same day, in the same rain. What do you think will happen to the seeds? Take 10 to 15 seconds of silent thinking time. Write one prediction.' WAIT TIME: 15 seconds, silently scanning the room. Then say: 'Turn to your partner and share. You have 2 minutes.' Walk around and listen without correcting. Cold-call 4 learners (2 who predict maize first, 2 who predict beans first) with: 'Tell us what you predicted and why.' Ask a follow-up to 2 of them: 'What makes you say that?' Record all predictions on the board under the headings 'Maize first', 'Beans first', 'Same time', 'Not sure'. Do not say which is correct. Say: 'Let us keep these predictions and test them against what we see.'	Think-Pair-Share (Predict-Observe-Explain, predict step): individual thinking first ensures every learner commits to an idea before seeing evidence. Pair talk lets learners rehearse and refine their language, and whole-class sharing reveals the range of ideas (including misconceptions) that the class will test against the videos.	Observation checklist: tick which learners write a prediction (not blank) and give a reason using 'because'. Listen for preconceptions such as 'big seeds always grow first', 'seeds need only water', 'seeds that look healthy must germinate', or 'plants grow only from the top'. Note 3 to 4 learners whose ideas to revisit. Indicator: at least 80% of learners have a written prediction with a reason before the video starts.
Observe Phase  VIDEO: Exponential growth & decay word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Root Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners watch the two time-lapse videos: 'Bean Time-Lapse - 25 days Soil cross section' and 'Growing corn time lapse #greentimelapse #gtl #timelapse'. They watch twice: first with no writing, just watching; second while jotting notes in a two-column table 'I notice' and 'I wonder'. They pay attention to: which seed splits or emerges first, the root growing down before or after the shoot grows up, how the stem becomes taller and thicker, how roots extend deeper (soil cross-section), branching or leaves appearing, and the seeds that do not sprout. If there is no connectivity, learners	Before playing: 'Watch carefully. The first time, just watch. The second time, write what you notice and what you wonder.' Play the bean video, pause at 3 points (for example when the root first appears, when the shoot emerges above the soil, when leaves open) and ask: 'What do you see happening right now?' WAIT TIME: 10 seconds, then cold-call 2 learners each time (6 learners total). Repeat for the maize video. Ask: 'What is the same in both plants? What is different?' WAIT TIME: 10 seconds, cold-call 3 learners. Prompt precision: 'Which part came out first, the root or the	Predict-Observe-Explain (observe step): watching the same phenomenon twice moves learners from general impressions to specific, recordable patterns. The 'I notice / I wonder' table separates evidence from questions, so that the later explanations and DQB questions are rooted in observations rather than guesses.	Collect 'I notice' and 'I wonder' tables from 5 learners at random. Look for observable indicators: at least 3 specific observations (e.g. 'the root came out before the shoot', 'the maize leaf was rolled', 'the bean stem was bent like a hook') and at least 2 wonderings. Oral probing: 'What did you see that surprised you?' Note the learners who only write general statements such as 'the plant grew' and prompt them to add details like the order and the part.

	observe the teacher's alternative: day-by-day pictures and a tray of real soaked maize and bean seeds (day 0), partly sprouted seeds (day 3 to 5) and seedlings (day 10 or more), passed around by group or viewed at the front. They compare the maize and bean sequences, sharing observations with their group.	shoot? How do you know?' If there is no power: hold up each picture, give 10 to 15 seconds for learners to look, narrate neutral facts only ('This is day 5'), and let learners name what they see. Pass the real seeds around in groups of 5 to 6 with the instruction 'Look, do not open yet'. Use simple English and write new words on the board with meanings: germination, emergence, shoot, root, stem.		
Explain Phase  VIDEO: Exponential growth & decay word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners work individually first, drawing and labelling their initial model. On an A4 page or notebook spread, they (1) draw a maize seed and a bean seed, (2) show what they think happens after planting, (3) show the root and shoot emerging, (4) show how the plant develops over time (for example Day 0, Day 5, Day 14, Day 25), and (5) add arrows and labels to explain their ideas, including what the seed gets from the soil and rain. They then complete the sentence starters: 'I think maize and bean plants grow differently because...', 'I notice that...', 'I wonder why...', and 'I think a seed needs _____ in order to germinate because...'. In their groups they use Think-Pair-Share to explain their model to a partner, and then each pair shares one idea with their group of 4 to 6. Learners add a different colour pen to mark ideas they borrow or change after discussion, so that their original thinking stays visible.	Say: 'You are scientists like Prof. Wangari Maathai, who observed trees, asked why, and tried to explain. Today you will show your first thinking. There is no right or wrong answer. I am not marking this.' Show a quick example on the board of how to label (arrow to root, label 'root'), using a non-maize, non-bean example such as a sukuma wiki seed, so you do not give the answer. Give 10 minutes for individual drawing. WAIT TIME: 10 to 15 seconds after each question you ask during the task. Circulate and ask 5 to 6 learners: 'Tell me about this arrow. What is happening here?' and 'Why did you draw the root there?' Do not correct; write down interesting ideas and misconceptions. Then say: 'Turn to your partner and explain your model in 1 minute each.' Cold-call 3 learners to share one idea from their partner's model (not their own) to build listening: 'What did your partner show that you had not thought about?' Finish with: 'Keep your first model safe. At the end of this unit you will come back and see how your thinking has changed.' Collect or store the models in folders marked with learner names. If many learners are in class, rotate	Model-based reasoning and Think-Pair-Share: drawing forces learners to make their thinking visible (what is inside the seed, what is needed, what changes over time). Explaining to a partner exposes gaps in the models. The unmarked baseline becomes the starting point for later model revision, so learners see learning as changing their own ideas.	Observation checklist for models: tick if the learner (a) draws both seeds, (b) shows the root and shoot emerging, (c) shows change over time, (d) uses at least 3 labels and arrows, and (e) gives a reason in the 'because' sentences. Record misconceptions such as 'the seed grows from the top only', 'the seed is dead until it is wet', 'the plant gets all its food from the soil', 'maize and beans grow the same way'. These are not marked. Indicator: at least 80% of the models have labelled arrows and a written reason, forming a baseline for the end-of-unit comparison.

		through the groups using the checklist rather than every individual.		
Driving Question Board (DQB) Creation  VIDEO: Exponential growth & decay word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  H5P: An A-maize-ing Story Source: Global Digital Library - Book Catalog  Search ARES for similar readings	Groups of 4 to 6 discuss: 'What questions do you have after watching the videos and considering the farmer's problem?' Each learner writes at least one question on a sticky note or a cut piece of paper (cards from old exercise books or manila paper work well), and the group chooses 2 to 3 of their best questions. Groups post their questions on the DQB (a chart or a wall section headed 'Why do maize and beans germinate, emerge and grow differently?'). Then, as a class, they sort the questions into clusters, for example: 'Inside the seed', 'Water, air, warmth', 'Seeds that did not germinate', 'How plants get taller and thicker', 'Roots and shoots', 'Why maize and beans differ'. They identify a question that appears many times, 'What is growth and development?', as the first to investigate in Lesson 2.	Display the driving question: 'What causes the differences in germination, emergence and growth of maize and bean plants?' Say: 'Now it is your turn to be the questioners. In your group, what questions do you have? Write one question per note. Start with Why, How, What or When.' Model one good question: 'Why did some seeds that looked healthy not germinate?' and one weak question and improve it: 'Is maize good?' becomes 'How does maize grow differently from beans?' Give 5 minutes for writing. WAIT TIME: 10 to 15 seconds after asking each group to choose their strongest questions. Circulate, and ask 2 groups: 'Can you turn this into a question that we can investigate?' Invite 6 groups (or all groups, if fewer) to read out and post one question each, then ask the class: 'Which questions seem to belong together?' Cold-call 3 learners to suggest the groupings. Point out the repeated question: 'Many of you asked what growth is, and what development is. Are they the same?' WAIT TIME: 10 seconds, cold-call 2 learners. Then say: 'This is our question for the next lesson: What is growth and development?' Record it clearly on the DQB. Leave space for 'What we have learned' and 'Evidence' columns to be added later. If space is limited, the DQB can be drawn on a sheet of manila paper or the back of a calendar.	Discussion and questioning with the DQB: learners' own questions give them ownership of the unit and a clear purpose for each future lesson. Clustering shows the links between questions (seed structure, conditions, hormones) and helps learners see that one big phenomenon can be broken into smaller investigable parts.	Check the DQB for observable indicators: each group has posted at least 2 questions, with at least one that is investigable (starts with how, why or what causes). Tally how many questions focus on (a) seed structure, (b) conditions for germination, (c) growth and development, (d) differences between maize and beans, to plan upcoming lessons. Oral probing: 'What would we need to do to answer this question?' Note groups that only ask yes/no questions and coach them to rephrase.
Model Building	Learners revisit their initial model and make a first, small update	Say: 'Models are never finished. Take your coloured pen and add	Model revision and Gallery Walk: adding evidence from the videos in	Exit ticket indicators: the learner names at least one specific piece

Phase  VIDEO: Exponential growth & decay word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  H5P: Tra's Red Bean Plant Source: Global Digital Library - Book Catalog  Search ARES for similar readings	using a different coloured pen: they add one thing they now notice from the videos that they did not draw before (for example the root emerging before the shoot, the way the maize shoot is covered by a sheath, or the curved bean stem), and one question mark next to a part of the model they are unsure about. They then do a Gallery Walk: groups place their models on desks or walls, and learners move around for about 6 minutes with sticky dots or ticks, leaving a comment such as 'I also thought...' or 'I wonder about...' on at least 2 other models. Finally they complete an exit ticket: 'What evidence did we gather today?' and 'How does this help answer our driving question?' They also record the new question: 'What is growth and development?' The finished models go into a folder as the 'Day 1 model' for revision at the end of the unit.	one thing you noticed from the videos. Put a question mark where you are not sure.' Give 5 minutes, WAIT TIME 10 to 15 seconds before taking questions. Set Gallery Walk rules: 'Walk quietly, read kindly, comment on two other models using 'I also thought...' or 'I wonder about...'. We do not say 'wrong' today.' Rotate groups every 2 minutes. After the walk, cold-call 3 learners: 'What idea did you see in another model that was different from yours?' Then summarise: 'Our models show many ideas: some of you think the seed needs water, some warmth, some air or soil. We will test these in the coming lessons.' Distribute exit tickets (half-sheets) and say: 'Write two sentences: What evidence did we gather today? How does this help answer our driving question?' Collect them at the door. Link forward: 'Next time we will begin by asking, what do we mean by growth, and how is it different from development?' Store the models and the DQB safely for the next lesson.	a different colour shows learners that models change when new evidence appears. Peer review exposes learners to alternative explanations, building the habit of comparing ideas, which will support Claim-Evidence-Reasoning explanations in later lessons.	of evidence from the videos (for example 'the root came out first in both seeds') and links it to the driving question in one sentence. Sort the tickets into three piles (clear link, partial link, no link) and note which learners need support with the term 'evidence'. Peer review: check that each learner left at least 2 comments on other models. Model check: look for an added feature in a second colour on at least 80% of the models.
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D. TEACHER REFLECTION

After the lesson, ask yourself: (1) Did every learner commit to a prediction and a first model before hearing others' ideas? (2) What were the 3 most common ideas or misconceptions about seeds, germination and growth (for example 'the seed is dead until it gets wet', 'only water is needed', 'plants grow like animals')? (3) Which questions on the DQB can be answered with the investigations planned in this unit, and which need to be reshaped? (4) Was the language manageable for learners, and which words need to be taught more clearly (germination, emergence, development)? (5) If the video was not available, did the pictures and real seeds create enough curiosity? (6) In large classes, were the groups and materials (sticky notes, manila paper, seeds) manageable, and who needs more support next lesson? Keep the baseline models, the exit tickets and a photograph or copy of the DQB for the end-of-unit revision.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In the time-lapse videos and pictures of maize and beans planted together after rain, we observed that the seeds germinated at different times, the root and shoot emerged, stems became taller and thicker, roots went deeper, some plants developed branches,
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	and some seeds did not germinate even though they looked healthy.
What did I learn?	We learned that the same rain and same planting day do not give the same result for every seed, and that we do not yet know what causes these differences. We drew our first models and learned that ideas can change when we find evidence. We also learned that we need to understand what growth and development mean.
How does this explain the phenomenon?	Our first explanation (not yet marked): maize and bean plants may grow differently because of differences in their seeds and in what each seed needs, such as water, air and warmth. We do not yet have evidence, so we have posted our questions on the Driving Question Board. Our next question is: What is growth and development?

LESSON 2 (80 minutes (double period) or 2 x 40 minutes): Meaning of Growth and Development; Structure of Monocot and Dicot Seeds

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners distinguish growth from development and relate the structures of maize (monocot) and bean (dicot) seeds to their functions, so they can begin to explain why seeds planted on the same day by a Kenyan farmer germinate, emerge and grow differently.
Knowledge	a) Distinguish between growth and development in living organisms. b) Relate structures of monocot and dicot seeds to their functions.
Skills	Observing and dissecting seeds; drawing and labelling biological diagrams; comparing and contrasting; communicating in clear scientific English; collaborating in small groups.
Attitudes	Unity: works harmoniously with others while examining seeds. Learning to Learn: reflects on own drawings and adjusts them after peer feedback. Appreciation of the importance of growth and development in plants.
Key Inquiry Question	How does growth occur in plants?
Purpose in Storyline	Lesson 1 presented the phenomenon: maize and beans planted together after rain behave differently. Lesson 2 establishes what growth and development mean and shows that maize and beans have different internal seed structures (food store in endosperm vs cotyledons). This gives a first possible reason for differences in emergence and early growth, and raises a new question: some seeds that look healthy do not germinate, so what is seed dormancy? That leads into Lesson 3.
Safety Notes	Use blunt scalpels or razor blades only under teacher supervision, or use a thumbnail/knife handled by one group member; cut away from the body on a tile or flat board. Soak seeds for 12-24 hours beforehand to soften them. Do not taste seeds, especially any treated with fungicide/pesticide (often pink or purple coated seed from agrovets). Wash hands after handling. Hand lenses only; never look at the sun.

B. LESSON OVERVIEW

Learners begin by recalling the time-lapse of the farmer's maize and beans after rainfall and predicting what is inside a maize grain and a bean seed. Using Think-Pair-Share they define growth (irreversible increase in size/dry mass) and development (changes in form and complexity, differentiation). In groups they then observe pre-soaked maize grains and bean seeds, draw and label them, and discuss the function of each structure with peers. A Gallery Walk lets groups compare drawings. Learners explain, using evidence, how the structure of each seed might affect early growth, add questions to the Driving Question Board (including 'What is seed dormancy?'), and revise their Day 1 model. Materials are local: maize grains, beans (rosecoco/Mwitmania/Wairimu), hand lenses or none, razor blades or knives, water, and printed or hand-drawn diagrams. No electricity or internet is needed; the time-lapse is recalled from Lesson 1 or replaced by the teacher's drawn sequence on the board.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO:	Learners recall the farmer who planted maize and beans on the	Display on the board (or recall orally) the time-lapse still images	Think-Pair-Share and Predict-Observe-Explain (the Predict	Observation checklist: Does the learner write a definition in their

No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: growth development plants monocot  EXERCISE: Embryo Growth and Development Practice Source: CK-12  Search ARES for similar readings	same day after rain. Individually they write in their books: (1) in their own words, what 'growth' and 'development' mean in the seedlings they saw, and (2) a quick sketch of what they think is inside a maize grain and a bean seed before germination. They pair up, compare, then share with the class. Learners also list one thing they predict is the same and one thing different between the two seeds.	from Lesson 1; if no projector, draw the sequence of a bean and a maize seedling. Say: 'Think quietly for 2 minutes. Write what growth means and what development means. Then sketch the inside of a maize grain and a bean.' Give WAIT TIME of 10-15 seconds after asking: 'Is a seedling getting taller the same as a seedling forming leaves?' Circulate and note ideas. Cold-call 4 pairs: 'Pair 3, what did you decide growth means?' Record all ideas on the board without correcting. Say: 'We will test these ideas with real seeds today.'	step). Individual thinking surfaces each learner's prior ideas before they see evidence; pairing lets them rehearse scientific language; recording predictions gives a baseline to compare with observations later in the lesson.	own words? Does the sketch show any internal parts (seed coat, embryo, food store)? Listen for misconceptions such as 'growth and development are the same thing', 'seeds are empty inside' or 'growth means only getting taller'. Tally how many of the 4 cold-called pairs mention increase in size versus change in form.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: growth development plants monocot  H5P: Tra's Red Bean Plant Source: Global Digital Library - Book Catalog  Search ARES for similar readings	In groups of 4-5 (sharing materials), learners receive soaked maize grains and soaked bean seeds. Step 1: Observe the outside with a hand lens or the naked eye, noting colour, shape, scar (hilum), small pore (micropyle) in beans, and the fused fruit wall and seed coat in maize. Step 2: Carefully split the bean along its curved edge and cut the maize grain lengthwise through the flat side. Step 3: Identify the testa, two cotyledons, plumule and radicle in the bean; pericarp/testa fused, endosperm, scutellum (single cotyledon), embryo (plumule with coleoptile, radicle with coleorhiza) in maize. Step 4: Draw large, labelled diagrams of both seeds with a title, magnification note and label lines. Step 5: Test for food stores by placing a drop of iodine solution (or diluted tincture of iodine from a pharmacy) on cut surfaces; starch turns blue-black, showing where food is stored (endosperm in maize, cotyledons in bean). Where iodine is unavailable, note the	Distribute pre-soaked seeds (prepared 12-24 hours earlier) and demonstrate the cut once, then say: 'Cut on the board, away from your fingers, one cutter per group.' Remind: 'Do not taste the seeds, and wash hands afterwards.' Ask while circulating: 'What do you see on the inside of the bean that you cannot see in the maize?' and 'Where do you think the baby plant is?' Give WAIT TIME of 10-15 seconds before taking responses. Cold-call 3 groups to point to the radicle and plumule on their specimen. Prompt struggling groups: 'Use the reading to match each part to its name.' Check drawings: 'Use a sharp pencil, single lines, no shading; label lines should not cross.' Use simple English and clarify terms: cotyledon = seed leaf, endosperm = food store tissue, plumule = shoot, radicle = root. For large classes, assign roles: cutter, reader, recorder, drawer, reporter, and rotate.	Predict-Observe-Explain (Observe step). Learners compare real structures with their earlier sketches, which creates cognitive conflict where predictions were wrong. Drawing and labelling forces careful attention to detail; roles support Unity by making each member accountable.	Observation checklist: Cuts the seed safely and correctly? Identifies at least 4 structures in each seed? Labels radicle and plumule correctly? Records one similarity and one difference between the seeds? Oral probing: 'Which part will become the root? Which will become the shoot?' Note misconceptions such as 'the cotyledon is the leaf of the plant above ground' or 'maize has no food store'.

	whitish floury texture of the endosperm versus the thick fleshy cotyledons. The reading provided by the teacher is used to confirm the names of parts. Learners link observations to the farmer's field: which seed parts will push out first and where the food comes from during emergence.			
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: growth development plants monocot  EXERCISE: Embryo Growth and Development Practice Source: CK-12  Search ARES for similar readings	Groups complete a comparison table of monocot (maize) and dicot (bean) seeds: number of cotyledons, location of food store, seed coat, protective sheaths, and function of each structure (e.g., testa protects the embryo, micropyle allows water entry, hilum shows the point of attachment, cotyledons store/absorb food, endosperm stores starch, scutellum absorbs digested food from the endosperm, coleoptile protects the plumule as it pushes through the soil, coleorhiza protects the radicle). Using Think-Pair-Share, learners discuss: 'How might the different food stores and protective structures affect how fast and how well maize and beans emerge from the soil?' Then they write a short Claim-Evidence-Reasoning statement, for example: Claim - maize and bean seeds are built differently; Evidence - our drawings show one cotyledon and a large endosperm in maize, two large cotyledons in bean; Reasoning - the position of the food store and protective sheaths affects how food reaches the embryo and how the shoot emerges. Learners also revisit their Predict definitions and rewrite them: growth = permanent increase in size, dry mass or cell number; development = changes in form and function, such as cell	Say: 'Turn to your partner and explain one function of a seed structure using evidence from your specimen.' Give WAIT TIME of 10-15 seconds, then cold-call 5 learners: 'Amina, what is the job of the coleoptile? What did you see that tells you this?' Press for evidence: 'How do you know?' Introduce the Gallery Walk: groups post drawings; each group visits 2 other displays with sticky notes for 'One thing that is clear' and 'One thing to improve'. Close the walk by asking: 'What did you see in another group's drawing that made you change your own?' Clarify: 'Growth and development often happen together but are not the same: a mandazi dough can increase in size but does not develop new organs.' Confirm the correct definitions only after learners have shared their reasoning.	Claim-Evidence-Reasoning, Think-Pair-Share and Gallery Walk. CER links observations to explanations; Gallery Walk exposes learners to alternative drawings and corrects errors through peer review, supporting Learning to Learn.	Peer review using a checklist: title present, label lines straight, all key parts labelled, correct function stated. Listen for CER quality: Does the claim match the evidence? Is the reasoning linked to function? Quick 3-question quiz (oral or written): (1) Name the part that becomes the root. (2) Where is food stored in maize? (3) Give one example of growth and one of development in beans. Record the number correct to identify who needs reteaching.

	differentiation and formation of leaves, roots and flowers. They give Kenyan examples: a maize seedling becoming taller (growth) and later forming a tassel and cobs (development).			
Driving Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: growth development plants monocot  EXERCISE: Embryo Growth and Development Practice Source: CK-12 Search ARES for similar readings	Learners return to the class Driving Question Board with the driving question: 'What causes the differences in germination, emergence and growth of maize and bean plants?' Each group writes on sticky notes (a) what they learned today about the seed structure and meaning of growth and development, and (b) new questions, such as 'Why do some healthy-looking seeds not germinate?', 'Does the seed coat stop water entering?', 'What is seed dormancy?'. They place notes under categories such as 'Seed structure', 'Conditions for germination', 'Dormancy', 'Growth'. Each learner completes an exit ticket: 'What evidence did we gather today?' and 'How does this help answer our driving question?'	Say: 'Look at the board. What do we now know that we did not know yesterday? What new question do you have?' Give WAIT TIME of 10-15 seconds. Cold-call 4 learners to read their questions aloud. Highlight the question 'What is seed dormancy?': 'Some of the farmer's seeds did not germinate even though they looked healthy. Could the seed itself be the reason? We will investigate this in our next lesson.' If no board space, use a manila paper chart or a chalkboard section. Collect exit tickets for review.	Discussion and questioning using the Driving Question Board. Linking each day's evidence to the central phenomenon helps learners see the unit as a coherent storyline and gives ownership of next steps through learner-generated questions.	Exit ticket success criteria: names at least one piece of evidence (e.g., 'maize has one cotyledon and endosperm; bean has two cotyledons') and links it to the driving question (e.g., 'different food stores may cause different emergence'). Check that at least two groups generate a question about why healthy seeds fail to germinate. Oral probing: 'Which of our questions can the seed structure alone answer, and which need more investigation?'
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: growth development plants monocot  EXERCISE: Embryo Growth and Development Practice	Learners take out their Day 1 drawing or explanation of why maize and beans behave differently. Individually they revise it, adding labelled diagrams of the seed structures and a short explanation of how growth and development begin from the embryo using the stored food. They add arrows showing food moving from endosperm (maize) or cotyledons (bean) to the embryo, and a question mark for 'unknown factors' (dormancy, water, oxygen, temperature) to be tested later. They write one sentence beginning 'Before I thought..., now I think...'	Say: 'Take out your Day 1 model. Using a different colour pen, add what you learned today. Show the seed parts and where the food comes from.' Give WAIT TIME of 10-15 seconds before asking: 'What did you change and why?' Cold-call 3 learners to share their changes. Point out gaps: 'Does your model explain why some healthy seeds did not germinate? What is missing?' Collect models for review. Assign optional homework: bring a labelled sketch of a seed from home (e.g., sukuma wiki, tomato, or kidney bean) and read about seed dormancy.	Model revision. Using a different colour pen makes learning visible and lets learners and the teacher see how understanding changes with evidence. The remaining gaps motivate the next lesson.	Model check: Does the revised model include correctly labelled seed structures? Does it show the stored food supplying the embryo? Does it identify at least one unanswered factor? Look for the 'Before I thought... now I think... because...' statement with an evidence-based reason. Provide feedback to those still confusing growth with development.

Source: CK-12 Search ARES for similar readings	because...'. Volunteers share revised models with a partner for a quick peer check.			
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D. TEACHER REFLECTION
Which learners still treat growth and development as the same thing, and which seed structures were most often mislabelled (e.g., coleoptile vs plumule)? Did groups share cutting tools and materials fairly, and was the soaking time sufficient for easy dissection? How well did learners use evidence from their own specimens rather than textbook memory in their CER statements? Which DQB questions show readiness for dormancy, and who needs extra support with scientific English terms before Lesson 3?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed and dissected soaked maize grains and bean seeds, and saw that maize has one cotyledon (scutellum) with a large endosperm and protective coleoptile and coleorhiza, while beans have two large cotyledons, a testa, hilum, micropyle, plumule and radicle.
What did I learn?	Growth is a permanent increase in size, dry mass or cell number, while development is the change in form and function as cells differentiate. Seed structures have specific functions such as protection, food storage, absorbing water and forming the new root and shoot.
How does this explain the phenomenon?	Maize and beans are built differently, so they store and supply food to the embryo in different ways, which may partly explain differences in emergence and early growth. Since some healthy-looking seeds still failed to germinate, other factors, such as seed dormancy, must also be investigated.

LESSON 3 (40 minutes (single period; the dormancy-breaking set-up is started in the lesson and observed over the following days)): Seed Dormancy: Why Do Some Healthy-Looking Seeds Refuse to Germinate?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners examine seed specimens to identify causes of seed dormancy and carry out simple activities to break dormancy, so that they can explain why some seeds planted on the same day after the same rain germinate quickly, slowly or not at all.
Knowledge	c) Outline causes and ways of breaking dormancy in seeds. Learners define seed dormancy and state its significance (survival through dry seasons, spreading germination over time). Learners outline causes: hard or impermeable seed coat, immature embryo, presence of chemical inhibitors, and absence of the right light or temperature cues. Learners outline ways of breaking dormancy: scarification (nicking, rubbing, soaking in warm water), leaching of inhibitors, stratification or chilling, and after-ripening.
Skills	Observing and examining seed specimens (hand lens); asking questions; planning and carrying out a simple investigation (NGSS SEP: Planning and Carrying Out Investigations); constructing explanations using Claim-Evidence-Reasoning (NGSS SEP: Constructing Explanations); communicating observations in simple, clear scientific English; collaborating in small groups.
Attitudes	Unity: works harmoniously with others while carrying out activities. Learning to Learn: reflects on own work and adjusts accordingly while carrying out activities to demonstrate ways of breaking seed dormancy. Appreciates that farmers in Kenya depend on understanding seed behaviour to plan planting and increase yields.
Key Inquiry Question	How does growth occur in plants? (Lesson 3 focus: What causes some seeds to remain dormant, and how can dormancy be broken so germination occurs?)
Purpose in Storyline	Lessons 1-2 established that seeds differ in structure (monocot maize versus dicot bean) and that growth begins from the embryo inside the seed. This lesson explains one reason why seeds planted together do not behave alike: some are dormant even though they look healthy. It answers part of the driving question (why some seeds fail or delay germination) and raises the next question: How does germination occur once dormancy is broken?
Safety Notes	Use only seeds not treated with pesticide (many commercial seed packets are dyed and chemically treated, so wash hands after handling, and never taste seeds). Use warm, not boiling, water, handled only by the teacher or a designated group member. Use sandpaper or a file on seeds held firmly on the bench, and keep blades and razors away from learners; nicking is done with a nail file or sandpaper. Wash hands after the practical and do not eat any seeds used in the experiment.

B. LESSON OVERVIEW

Return to the anchoring phenomenon: a farmer in Kitale plants maize and beans on the same day after the first rains. Most germinate in a few days, some take longer, and some healthy-looking seeds never sprout. Learners predict why, then examine seed specimens (maize, beans, and other locally available seeds such as cowpeas, pigeon peas, Acacia or Croton seeds, and avocado or pawpaw seeds if available) to look for features that might block germination. They watch the Khan Academy video 'Seed dormancy | Plant growth and development | Biology' where connectivity allows, or use the teacher-prepared alternative (a teacher-read passage with a labelled chalkboard diagram). In small groups they set up simple dormancy-breaking treatments (control, rubbing with sandpaper, warm-water soak, and cold or leaching treatment) using locally available materials. They use Claim-Evidence-Reasoning to explain what causes dormancy and add evidence to the Driving Question Board, ending with the new question: How does germination occur?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Critically looking at data on ROC and economic growth over millenia Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the phenomenon: maize and beans planted together after rain in Trans Nzoia, yet some seeds germinate within days, some take longer, and some healthy-looking ones never germinate. Using Think-Pair-Share, each learner writes a prediction in their notebook: 'Why might a healthy-looking seed fail to germinate even when it has rain? What could we do to make it germinate?' Learners sketch one seed and mark where they think the problem might be (seed coat, embryo, inside the seed). They pair up and then share with the class.	Display on the board: 'Same day. Same rain. Different results. Why?' Say: 'Think quietly for one minute. Write your prediction and draw the seed with an arrow to the part you think is responsible.' WAIT TIME: 10-15 seconds of silence after posing the question before accepting any answers. Move around and look at the drawings. Say: 'Turn to your partner and explain your idea in simple English. Partner, ask: What makes you think so?' Cold-call 4 pairs, including at least 2 who have not spoken in the previous lessons. Record all predictions on the board under the heading 'Our predictions' without judging them (for example: 'seed is dead', 'no water', 'seed coat is too hard', 'seed is still sleeping'). Say: 'Today we will test these ideas with real seeds.' Introduce the word 'dormancy' as 'a resting state where a living seed does not germinate even when conditions look right'.	Think-Pair-Share and Predict-Observe-Explain (the Predict step): Individual thinking commits each learner to an idea before evidence is seen, pair talk lets learners rehearse scientific English, and recording predictions publicly creates ideas the class can later confirm or revise. This ties directly to the anchoring phenomenon.	Observable indicators: (1) Each learner's notebook has a written prediction and an annotated seed drawing. (2) Count learners who locate the cause at the seed coat, the embryo, or only 'lack of water'. (3) Listen for misconceptions such as 'a seed that does not germinate is dead' or 'all seeds germinate immediately if watered'. Tally these on a quick checklist for use in the Explain phase.
Observe Phase  VIDEO: Critically looking at data on ROC and economic growth over millenia Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seed Dormancy and Germination	Part A (10 min): In groups of 4-5 (sharing hand lenses and specimens), learners examine seed specimens: maize grains, bean seeds, cowpeas, and any hard-coated local seeds available (for example Acacia, Senna/Cassia, Croton, Grevillea or avocado seeds). They feel the hardness, observe the seed coat with a hand lens, and compare the coat thickness of a dry seed with a seed that has been soaked. They record observations in a table: seed type, coat description (thin/thick, smooth/hard), water	Before the lesson, prepare per group: seed samples in small cups, one hand lens (or share between two groups), sandpaper strips, warm water in a flask, cotton wool or cloth, 4 containers (cut plastic bottles), labels, and pencils. If connectivity is available, play the Khan Academy video 'Seed dormancy Plant growth and development Biology' once on a projector or the teacher's phone with the class gathered around it; pause twice to ask 'What is stopping this seed from germinating?' with WAIT TIME of	Predict-Observe-Explain (the Observe step): Learners collect firsthand evidence about seed coats, water uptake and treatment effects, which they will later compare with their predictions. The control group enables fair comparison, and group roles and shared materials build Unity while keeping the practical feasible for large classes.	Use an observation checklist per group with these indicators: (1) handles seeds safely and correctly uses the hand lens; (2) records at least two observable seed-coat features; (3) sets up all four containers correctly with labels and a control; (4) every member has a role and takes part; (5) uses precise words such as 'hard', 'swollen', 'absorbed' in simple scientific English. Use oral probes: 'How do you know the water entered the seed?' and 'Why must the amount of water be the same in all containers?' to identify

- Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	uptake after soaking (swells or not). They then watch the Khan Academy video on seed dormancy (or hear the teacher-read alternative) and note causes of dormancy. Part B (10 min): Groups carry out activities to demonstrate ways of breaking seed dormancy. Each group sets up 4 labelled containers (cut plastic bottles or clay saucers lined with wet cotton wool, cloth or paper towel) with 5-6 hard-coated seeds each (for example Senna, Acacia, or old bean seeds): (1) control - untreated seeds, (2) scarification - seed coat rubbed lightly on sandpaper or a rough stone, (3) warm-water soak - seeds soaked in warm (not boiling) water for a few hours, (4) leaching/chilling - seeds washed in several changes of water or kept in a cool place. All containers are kept moist and placed in the same location, and groups record observations daily.	10-15 seconds before taking responses. If there is no connectivity, use the teacher-prepared alternative: read aloud a 150-word passage and draw a labelled diagram on the board showing a hard seed coat, inhibitors, and an immature embryo. Say: 'Hold the seed. Is the coat hard? Does water enter? Write what you SEE, not what you think.' Circulate and ask each group: 'Which seed took up water fastest? Why do you think so?' and 'What is your control and why do we need it?' Cold-call 3 groups to report one observation each. Remind groups: 'Warm water only, not boiling, and wash hands after handling seeds.' Assign roles (materials manager, recorder, timekeeper, reporter) to support unity and fair participation. Where the class is large, groups can share sandpaper and flasks.		misconceptions.
Explain Phase  VIDEO: Critically looking at data on ROC and economic growth over millenia Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	Learners use Claim-Evidence-Reasoning (CER) to answer: 'What causes seed dormancy, and how can it be broken?' On a sheet or in their notebooks, each group writes a Claim (for example 'A hard seed coat prevents water from entering, so the seed remains dormant'), supports it with Evidence from the seed examinations and the video (for example 'The Senna seeds did not swell after soaking, but the rubbed seeds did'), and gives Reasoning linking it to biology (water uptake, called imbibition, activates enzymes in the embryo; the coat blocks water and gas entry; inhibitors must be removed or broken down). Learners then link causes of dormancy to the farmer's field: why might some	Say: 'Use the CER frame: My claim is... My evidence is... This is because...' Display the frame on the board. Pose: 'Why did the control seeds behave differently from the rubbed seeds?' WAIT TIME: 10-15 seconds before calling on anyone. Cold-call 4 groups to share their CER aloud, and ask a different learner in the group to defend it, not the recorder. Probe: 'What other cause of dormancy could explain a seed that is healthy but still does not germinate?' (guide towards inhibitors, immature embryo, and temperature/light cues). Clarify tricky terms on the board: dormancy, scarification, imbibition, inhibitor, embryo. Connect to the phenomenon: 'How could this	Claim-Evidence-Reasoning (CER): Learners are required to connect each claim to evidence from their own observations and to scientific reasoning, moving from describing seeds to explaining why they behave differently. Discussion and questioning reveal gaps and confirm correct ideas, and linking back to the farmer's field makes the explanation relevant to the phenomenon.	Use a CER check with these indicators: (1) the claim is a clear statement about the cause of dormancy; (2) at least one piece of evidence comes from the class observations, not only the video; (3) the reasoning mentions water or gas entry, inhibitors, or embryo development; (4) learners correct the misconception that non-germinating seeds are always dead. Peer-review CER statements by swapping with another group and giving one star (good point) and one wish (improvement). The teacher notes groups needing reteaching.

	bean seeds delay in the soil? They also discuss the significance of dormancy for plants in Kenyan ecosystems, such as seeds of Acacia in the Tsavo drylands waiting for adequate rain.	explain why some of the farmer's seeds did not germinate after the same rain?' Address misconceptions flagged in the Predict phase, for example 'Not germinating does not always mean dead; the seed may be dormant.' Summarise the causes (hard seed coat, chemical inhibitors, immature embryo, environmental cues) and the ways of breaking dormancy (scarification, soaking, leaching, chilling/stratification, after-ripening), and mention farmers' practices such as soaking bean or pigeon pea seeds overnight, or cracking hard seeds slightly before planting.		
Driving Question Board (DQB) Creation  VIDEO: Critically looking at data on ROC and economic growth over millenia Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners revisit the class Driving Question Board: 'What causes the differences in germination, emergence and growth of maize and bean plants?' In their groups they write new sticky notes or strips of paper with (a) one thing they now know (for example 'Hard seed coats can block water and cause dormancy') and (b) one new question (for example 'What happens inside the seed once water enters?' or 'How does the seed know when to start growing?'). They post these on the board and group them under headings such as 'Causes of delay or failure' and 'What happens during germination'. Learners also move any earlier question that has now been answered into an 'Answered' section.	Say: 'Walk to the board with one note that shows what we have learned today and one that shows what we still wonder.' Allow WAIT TIME of 10-15 seconds for learners to think before writing. Cold-call 3 learners to read their notes aloud and justify where they placed them. Highlight the question 'How does germination occur?' and say: 'This is our next step. Once dormancy is broken, what exactly happens in the seed?' Ask: 'Which of our earlier questions about maize and beans can we now partly answer?' Move 2-3 questions to the 'Answered' section with learners' agreement, citing the evidence. Keep the DQB visible in the classroom, and remind learners to watch the dormancy set-ups daily.	Discussion and questioning using the Driving Question Board: Learners see how each lesson's evidence contributes to the main question, track what is answered or unanswered, and generate the next investigation question themselves, so learning is purposeful and connected.	Observable indicators: (1) each group posts at least one 'know' note and one 'wonder' note; (2) notes use evidence from today's work, not just recall; (3) at least one question leads naturally to germination (for example the process, the role of water, oxygen or temperature); (4) learners can explain why they placed a note in a certain category. Record the quality of questions on the observation checklist.
Model Building Phase  VIDEO: Critically looking	Learners return to their Day 1 drawing and explanation of why the farmer's seeds behave differently. Individually, they revise it by adding a labelled sketch of a	Say: 'Take out your Day 1 model. Using a different colour pen, add what you have learned about dormancy. Show cause and solution.' WAIT TIME: 10-15	Model revision: Learners compare their earlier thinking with new evidence and make visible changes to their model, showing conceptual growth. Peer review	Model check indicators: (1) the revised drawing includes a labelled seed coat, embryo and inhibitors; (2) at least one cause of dormancy and one method of breaking it are

at data on ROC and economic growth over millennia Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	dormant seed showing the hard seed coat, inhibitors, and embryo, and arrows showing how scarification or soaking allows water to enter. They add a short note: 'New idea: Some seeds fail to germinate because _____, and farmers can help by _____. ' In pairs, they swap drawings and give one piece of peer feedback. Before leaving, each learner completes an exit ticket answering: 'What evidence did we gather today?' and 'How does this help answer our driving question?'	seconds after giving the instruction before circulating, so learners start thinking independently. Walk around and ask 3-4 learners: 'What did you add and what evidence made you change it?' Cold-call 2 pairs to share a revision and the evidence behind it. Say: 'Place your revised model in your portfolio. Over the next few days we will check our containers and add more evidence.' Hand out or write the exit ticket questions on the board and collect them at the door. Remind groups to keep their dormancy set-ups in a safe, shaded spot, to keep the cotton or cloth moist, and to record daily observations (this supports the later Gallery Walk of germination results). Announce the next lesson: 'How does germination occur?'	and exit tickets strengthen reflection (Learning to Learn) and show how today's evidence connects to the driving question.	shown with arrows or notes; (3) a change from the Day 1 model is clearly visible; (4) the exit ticket cites specific evidence (for example 'rubbed seeds swelled but untreated seeds did not') and links it to why some seeds in the farmer's field did not germinate. Use a quick quiz of 3 questions at the start of the next lesson on causes and ways of breaking dormancy.
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D. TEACHER REFLECTION

After the lesson, consider: Which misconceptions about non-germinating seeds persisted (for example 'dormant means dead')? Were group roles effective in ensuring that every learner handled seeds and contributed ideas? Did the large class size limit hand-lens use or sandpaper sharing, and how can materials be organised better? Did learners use the CER structure with real evidence from their containers, or mostly the video? Was the teacher-prepared alternative needed, and was it as clear as the video? What indicators on the checklist show that learners are ready for the germination lesson? Plan to check the set-ups over the next days and use the results in the Gallery Walk.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We examined maize, bean and other local seeds with hand lenses, noting hard and thin seed coats and how much water the seeds absorbed. We set up control, scarified, warm-water-soaked and leached/chilled seeds and recorded changes over time.
What did I learn?	Seed dormancy is a resting state in which a living seed fails to germinate even when conditions seem suitable. Causes include a hard impermeable seed coat, chemical inhibitors, an immature embryo, and unsuitable environmental cues. Dormancy can be broken by scarification, soaking in warm water, leaching, chilling (stratification) and after-ripening. Dormancy helps seeds survive unfavourable seasons and spreads germination over time.
How does this explain the phenomenon?	Some seeds in the farmer's field do not germinate or are delayed even after the same rain because they are dormant: their hard seed coats may block water entry, inhibitors may need to be washed away or broken down, or their embryos may not be ready. Evidence: in our containers, treated seeds took up water and responded sooner than untreated controls. This explains part of the

	difference in germination between maize and beans, but we still need to find out how germination occurs once dormancy is broken.
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LESSON 4 (80 minutes (double period); set-up in lesson, observations continue over 5-7 days in 5-minute slots): Germination: Process, Types and Conditions Necessary for Germination in Maize and Bean Seeds

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate the conditions necessary for germination and the types of germination (hypogeal and epigeal) using maize and bean seeds, so they can explain why seeds planted on the same day after the same rain germinate and emerge differently.
Knowledge	Learners will be able to: (d) investigate conditions necessary for seed germination, and (e) investigate types of germination in plants. They outline the process of germination (water uptake, activation of enzymes, radicle emergence, then plumule), and distinguish hypogeal germination (maize, which keeps cotyledons below the soil) from epigeal germination (common bean, where cotyledons are lifted above the soil).
Skills	Predicting, planning a fair test and identifying variables (SEP: Planning and Carrying Out Investigations); recording and comparing observations; constructing explanations using Claim-Evidence-Reasoning (SEP: Constructing Explanations); communicating in clear scientific English. Core competency: Learning to Learn, where the learner reflects on their own work and adjusts accordingly.
Attitudes	Unity: the learner works harmoniously with others while carrying out experiments to investigate conditions necessary for seed germination. Curiosity and respect for evidence when results differ from predictions. Appreciation of the importance of growth and development in plants (outcome h).
Key Inquiry Question	How does growth occur in plants? (Lesson focus: What conditions and processes must be in place before a seed can begin to grow, and how do maize and beans differ as they emerge?)
Purpose in Storyline	Lessons 1-3 established seed structure and dormancy. Lesson 4 gathers evidence for the first part of the phenomenon: why seeds planted together after the same rain germinate at different times, or fail even when they look healthy. Learners find that water alone is not enough, since oxygen, suitable temperature and a viable, non-dormant seed are also needed, and that maize and beans emerge differently. This raises the next question: once the seedling has emerged, How do plants show growth?
Safety Notes	Use only clean, untreated seeds. Do NOT use seed maize or bean seed treated with pesticide (often pink or red dyed from agrovets). If treated seed is the only supply, the teacher handles it and learners wear gloves, and it is not used for eating. Wash hands after handling seeds, soil and wet cotton or tissue. Do not taste any seeds. Boil water only under teacher control, then cool it fully before use. Keep a heat source away from learner benches. Clear spilt water immediately to prevent slips. Dispose of mouldy seeds in a sealed bag, and learners must not sniff them closely.

B. LESSON OVERVIEW

Learners return to the anchoring phenomenon of the farmer who plants maize and beans on the same day after rain, yet sees uneven germination, delayed emergence and some failed seeds. In small groups of 5-6 (suited to large classes) they predict, then set up a fair test using locally available materials (plastic cups, old jam jars, cotton wool or tissue, sawdust or soil, polythene bags, a sufuria, a cold place such as a wet cloth in a shaded clay pot or fridge if available): Set A with water, air and warmth (control); Set B dry seeds; Set C seeds submerged in boiled, cooled water with oil on top (little oxygen); Set D cold or cool conditions; and Set E boiled (non-viable) seeds. Maize and bean seeds are used in each set. Set-up takes about 30 minutes and groups observe for 5-7 days. To gather evidence on the same day, learners also examine pre-germinated seeds (teacher-prepared 4-6 days before) of maize and beans, drawing and comparing hypogeal and epigeal germination. If a time-lapse video is available it is shown on a phone or projector; otherwise the teacher uses a prepared picture sequence or flip chart. Learners use Predict-Observe-Explain, Think-Pair-Share and Claim-Evidence-Reasoning, then add questions to the Driving Question Board and revise their Day 1 model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Resources and population growth Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  H5P: Tra's Red Bean Plant Source: Global Digital Library - Book Catalog  Search ARES for similar readings	Learners recall the phenomenon: a farmer in Kitale plants maize and beans on the same day after the first rains; some seeds sprout in 4 days, some in 10, and some healthy-looking seeds never sprout. Individually, they write a prediction in their notebooks: 'Which conditions does a seed need to germinate? What will happen to maize and bean seeds in: (a) dry air, (b) water with no air, (c) cold, (d) warm, moist, open air?' They sketch what they expect to see after 5 days and give one reason. They then pair up (Think-Pair-Share) and compare. Finally, groups agree on a group prediction chart for the 5 set-ups, listing the variable they will change and the ones they must keep the same.	Display or read the phenomenon: 'A farmer in Kitale plants maize and beans on the same day, under the same rain. Why might some seeds sprout quickly, some slowly and some not at all?' Say: 'Think quietly and write your own prediction. You have 2 minutes. No talking yet.' Then WAIT TIME 10-15 seconds after asking 'What does a seed need before it can start growing?' before taking any answer. Say: 'Turn to your partner. Share your prediction and tell them WHY you think so.' Circulate and listen for misconceptions such as 'seeds need soil to germinate' or 'seeds need light to germinate'. Cold-call 4 pairs: 'Amina and Otieno, what did your partner predict, and what was the reason?' Record all predictions on the board without judging. Ask: 'Which prediction could we test? How would we make the test fair?'	Predict-Observe-Explain (Predict step) and Think-Pair-Share. Committing to a written prediction before any evidence makes learners' existing ideas visible and gives them something to confront with data later. Think-Pair-Share lets every learner, including quiet ones in a large class, rehearse an explanation before speaking publicly, which builds confidence and surfaces competing ideas for investigation.	Observation checklist: (1) learner writes a prediction with a reason, not just a guess; (2) learner mentions at least one condition (water, air or oxygen, warmth); (3) learner can state what would be kept the same. Oral probing: 'Why do you think soil is needed?' to identify the misconception that soil, light or fertiliser is essential for germination. Teacher tallies on the board how many pairs predicted that each condition is necessary. Learners who give only 'water' are noted for follow-up questions.
Observe Phase  VIDEO: Organism growth and the environment Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seed Dormancy and Germination - Advanced (Flexbook)	Part A (about 30 min): Groups set up the fair test. Each of 5 labelled containers receives 5 maize and 5 bean seeds on moist cotton wool or tissue: A = moist, open to air, room temperature (control); B = dry cotton, no water; C = seeds fully submerged in boiled and cooled water with a thin layer of cooking oil on top (little oxygen); D = moist, in a cool place (wet cloth in a shaded clay pot, or a fridge if the school has one); E = boiled-then-cooled seeds on moist cotton (non-viable seeds). Groups label, draw the set-up and make a	Give out materials in advance in trays for each group (one tray per 5-6 learners). Say: 'Roles today: materials manager, recorder, timekeeper, reporter, and safety officer. Everyone handles the seeds at least once.' Demonstrate one set-up for 2 minutes, then WAIT TIME 10-15 seconds after asking: 'Which condition are we changing in Container C, and which ones must stay the same?' Cold-call 3 groups to state their variable. Circulate with prompts: 'What are you measuring? How will you count a seed as	Predict-Observe-Explain (Observe step). Learners test their predictions with controlled variables, so they see that any difference in germination can be linked to one condition. Comparing maize and bean seedlings side by side helps them describe types of germination from evidence rather than memory, and connects directly to the phenomenon of why the two crops emerge differently.	Observation checklist with indicators: (1) group labels every container and identifies the changed variable; (2) all members handle materials and take turns; (3) measurements are recorded with units (mm) and dates; (4) drawings show correct labelled parts (radicle, plumule, cotyledon, coleoptile, hypocotyl); (5) learner correctly tells which seed is hypogeal and which is epigeal. Oral probe: 'Why do we include dry seeds and boiled seeds?' Note groups that confuse 'germination' with 'growth' for follow-up.

Source: CK-12 Search ARES for similar readings	results table (day, number germinated, radicle length in mm, notes). Part B (about 25 min): Groups examine teacher-prepared germinating maize and bean seedlings (started 4-6 days earlier) and, if available, watch a 2-minute time-lapse. They draw and label both: maize with the coleoptile and plumule emerging while the cotyledon stays below ground (hypogeal), and bean with the hypocotyl arching and lifting the cotyledons above the soil (epigeal). Part C (5-7 days): Groups record observations in 5-minute daily slots.	germinated?' 'Why are we using 5 seeds of each type instead of 1?' For Part B, ask: 'Where is the cotyledon in the maize seedling? Where is it in the bean?' WAIT 10-15 seconds before accepting answers. If no electricity or data, hold up a prepared chart or sketch sequence of the time-lapse. Teacher prepares a spare container of already-germinated seeds in case some groups' set-ups fail. Remind: 'Unity: share equipment fairly and listen to every group member's ideas.'		
Explain Phase  VIDEO: Photosynthesis in organisms Source: Khan Academy (English - US curriculum) Search ARES for similar videos  H5P: Tra's Red Bean Plant Source: Global Digital Library - Book Catalog Search ARES for similar readings	Learners meet after the first results (day 3-5) and use Claim-Evidence-Reasoning to answer: 'What conditions are necessary for maize and bean seeds to germinate, and how do the two seed types differ in emergence?' Each group writes a claim, cites evidence from its table (e.g., 'In A, 9 of 10 seeds germinated by day 4; in B, 0 of 10'), and gives reasoning in terms of biology: water activates enzymes and softens the seed coat, oxygen is needed for aerobic respiration to release energy, and suitable temperature allows enzymes to work. They explain why boiled seeds failed (embryo dead, so not viable) and link back to the phenomenon: healthy-looking seeds that failed could be dead, dormant, waterlogged (low oxygen) or too cold or dry. They then hold a Gallery Walk: groups post their CER and labelled drawings, and visit other groups' work to give one 'star' (strength) and one 'wish' (suggestion).	Say: 'Open your results tables. Write your claim first, then your evidence, then your reasoning. Use the sentence starter: I claim that ... because the data show ... which happens because ...'. Ask: 'In Container C there was water but few seeds germinated. What does that tell you?' WAIT TIME 10-15 seconds, then cold-call 3 learners. Press for reasoning: 'Which biological process needs oxygen? Why would waterlogged shamba soils in Kisumu affect maize seed?' Connect: 'Why might the farmer's seeds in the dry patch of the field fail, and the ones in a flooded patch near Lake Victoria also fail?' Prompt on dormancy: 'Some seeds were in good conditions but didn't germinate. What from Lesson 3 could explain this?' For the Gallery Walk, give 8 minutes and a timer: 'Move clockwise every 2 minutes. Leave one star and one wish on a sticky note or a piece of paper.' Clarify terms in simple English: 'hypogeal means below ground; epigeal means above ground.' Correct	Claim-Evidence-Reasoning with a Gallery Walk. CER forces learners to separate what they observed from the scientific explanation, reducing guessing. Gallery Walk exposes them to other groups' data, including results that may differ, which teaches that evidence from many trials strengthens a claim and encourages peer review.	Use a CER rubric: Claim (clear, testable, matches question); Evidence (specific numbers or observations from the table); Reasoning (links evidence to a biological principle such as respiration, enzymes or water uptake). Exit-level indicators: learner states at least 3 conditions (water, oxygen, suitable temperature) and one internal factor (viable, non-dormant seed); learner correctly classifies maize as hypogeal and bean as epigeal with a structural reason. Peer review: stars and wishes are collected and checked by the teacher. Oral probing identifies the misconception that 'seeds need light or soil to germinate.'

		misconceptions with evidence, not telling.		
Driving Question Board (DQB) Creation  VIDEO: Linear vs. exponential growth: from data (example 2) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  H5P: Tra's Red Bean Plant Source: Global Digital Library - Book Catalog  Search ARES for similar readings	Learners return to the Driving Question Board with the main question: 'What causes the differences in germination, emergence and growth of maize and bean plants?' In groups, they write on sticky notes or paper strips: (1) one thing we now know about germination (with evidence), and (2) one new question. Examples: 'Why do bean cotyledons rise but maize ones stay?', 'Why did some seeds in A germinate later than others?', 'How do plants show growth after the seedling emerges?' They place notes under headings: Germination conditions, Types of germination, Growth after emergence, Unanswered. They complete an exit ticket: 'What evidence did we gather today?' and 'How does this help answer our driving question?'	Say: 'Look at the board. Which of today's findings help answer the driving question? Which parts are still a mystery?' After asking 'Which question on the board can we answer next?' give WAIT TIME 10-15 seconds. Cold-call 4 groups to read one new question and say where to place it. Group questions with learners, highlighting the question: 'Our seedlings have emerged, but how do plants show growth? How would we measure it?' Tell learners: 'That is our focus in the next lesson.' Collect exit tickets as learners leave. If no stickies are available, use torn exercise-book paper and masking tape or chalk-drawn columns on the blackboard.	Discussion and questioning using the Driving Question Board. Linking each new finding to the central question keeps learning coherent, shows how evidence builds toward an explanation, and turns learners' unanswered questions into the direction of the unit. Learners see that science advances through questions generated by evidence.	Exit ticket indicators: (1) names specific evidence such as 'Seeds in dry cotton did not germinate'; (2) links the evidence to the driving question; (3) writes at least one productive new question. Teacher sorts tickets into three piles: secure, partial, and misconception (e.g., 'seeds need sunlight to germinate') and plans a quick re-teach at the start of Lesson 5. Observe DQB placements for accuracy.
Model Building Phase  VIDEO: Resources and population growth Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  H5P: Tra's Red Bean Plant Source: Global Digital Library - Book Catalog  Search ARES for similar readings	Learners take out their Day 1 drawings and explanations of the phenomenon. Individually, they revise the model by adding: (a) the conditions needed to germinate (water, oxygen, suitable temperature, viable and non-dormant seed) shown with arrows; (b) a labelled diagram of maize (hypogeal) and bean (epigeal) germination; and (c) a short note answering: 'Why might the farmer's seeds, planted together after the same rain, germinate at different times or fail?' They use a different colour pen for revisions so that changes are visible. They swap with a partner for peer review using the prompt: 'What evidence supports each new part of the	Say: 'Take your Day 1 model. Use a different colour to add what you now know. Every addition should be backed by evidence from our containers.' Circulate and ask: 'Where did you get this idea? Which container shows it?' WAIT TIME 10-15 seconds before following up. Cold-call 3 learners to describe one change they made and why. Pair review for 5 minutes: 'Give your partner one suggestion and one strength.' Say: 'Keep your model safe; we will add growth next.' Collect a sample of 5-6 models for checking. Remind groups to observe containers for the remaining days and record data.	Model revision. Revisiting the Day 1 model with a new colour makes learning visible to the learner and the teacher. Linking each change to evidence shows how scientific models are improved, and it prompts learners to identify gaps (growth after emergence) that the next investigations will fill.	Model check indicators: (1) at least three conditions for germination are added with arrows or labels; (2) hypogeal and epigeal types are drawn and labelled correctly (cotyledons below or above soil); (3) revisions are linked to evidence; (4) the explanation addresses why seeds planted together behave differently (viability, dormancy, uneven moisture, oxygen or temperature); (5) peer feedback is specific. Note learners who still hold misconceptions for individual conferencing.

	model? What is still missing?' They leave a space labelled 'What happens after emergence?' to complete in Lesson 5.			
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D. TEACHER REFLECTION

After the lesson, consider: Did learners distinguish between germination and growth? Which groups could name the variable they changed and the ones kept constant? Did any learners still believe that seeds need soil or light to germinate, and what evidence changed their minds? Were the materials sufficient for the class size, and did group roles ensure everyone participated? Were the seed set-ups reliable (check for mould, drying out, or treated seed)? Did the pre-germinated seedlings arrive on time? How well did learners use CER with specific numbers? What adjustments are needed before Lesson 5 to measure growth, such as preparing extra seedlings, rulers and string, and a recording table? Also record which learners need extra support with English scientific vocabulary.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We set up maize and bean seeds in containers with different conditions (moist and open to air, dry, submerged with little oxygen, cool, and boiled seeds) and compared them over several days. We also examined and drew germinating maize and bean seedlings. Record the number of seeds germinated in each container and the differences between how the maize and bean seedlings emerged.
What did I learn?	Seeds need water, oxygen and a suitable temperature to germinate, and they must be alive (viable) and not dormant. Germination starts with water uptake, then the radicle emerges, then the plumule. Maize shows hypogeal germination (cotyledon stays below ground), while bean shows epigeal germination (cotyledons are lifted above ground).
How does this explain the phenomenon?	Seeds planted on the same day after the same rain can germinate at different times, or fail, because of differences in moisture, oxygen (for example in waterlogged soil), temperature, seed viability and dormancy. Maize and beans also emerge differently because of their type of germination. After emergence, how do plants show growth? This will be investigated in the next lesson.

LESSON 5 (80 minutes (double period) – can be split into 40 + 40 minutes): Primary and Secondary Growth: How Do Plants Show Growth?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather evidence from stem sections and models to distinguish primary growth (increase in length) from secondary growth (increase in girth). They link this to the maize and bean seedlings that grew taller and thicker after germination, and to the observation that bean plants branch and thicken while maize does not.
Knowledge	By the end of the lesson the learner should be able to: (1) illustrate primary and secondary growth in flowering plants (SLO f); (2) relate the structures seen in stem sections to their functions (xylem, phloem, cambium, cortex, epidermis); (3) explain why a dicot stem such as bean or a Grevillea/Cypress tree thickens while a monocot stem such as maize mostly does not; (4) connect this to the germination types compared in earlier lessons (hypogeal in maize, epigeal in bean).
Skills	Observing prepared slides or photomicrographs under a light microscope; drawing and labelling stem sections with a title, scale and annotations; using models to represent stages of secondary growth; Claim-Evidence-Reasoning writing; peer review; working safely in groups. NGSS SEP: Developing and Using Models; Analyzing and Interpreting Data; Constructing Explanations.
Attitudes	Unity: works harmoniously with others while sharing microscopes and materials. Learning to Learn: reflects on own drawings and models and adjusts them using peer feedback and evidence. Appreciation of the importance of growth and development in plants, including trees in school nurseries and for preventing global warming.
Key Inquiry Question	How does growth occur in plants? (Question raised for this lesson: How do plants show growth?)
Purpose in Storyline	Lessons 1–4 established seed structure, dormancy, and the conditions for germination, and compared hypogeal and epigeal germination. Learners now look beyond germination to the growth that follows: why seedlings get taller and thicker, why roots go deeper, and why beans branch. This lesson adds the explanation of growth in length and girth through meristems. It ends with the new question 'How can we measure growth in plants?', leading to Lesson 6.
Safety Notes	Handle microscope slides and cover slips carefully; report broken glass to the teacher and do not pick it up with bare hands. Carry microscopes with two hands. Use a sharp razor blade or knife only under teacher supervision, cutting away from the body on a firm board. Wash hands after handling plant material and soil. If microscopes are few, rotate groups of 6 at stations and use photomicrographs or teacher-drawn charts at other stations. Cutting stems are used in place of electricity-dependent resources.

B. LESSON OVERVIEW

The anchoring phenomenon showed maize and bean seeds planted on the same day after rain, yet growing differently: stems getting taller and thicker, roots going deeper, and some plants branching. Today learners focus on how this growth happens. They predict which parts of a young maize and bean seedling and which parts of an old tree such as a Grevillea or Eucalyptus grow in length or girth, then observe stem sections (prepared slides, photomicrographs or hand-cut sections of young bean and maize stems) and use the reading on primary and secondary growth. They draw and label sections, build a simple plasticine or paper model of secondary growth in a woody dicot stem, and explain using Claim-Evidence-Reasoning. They add questions to the Driving Question Board and revise their class model of why maize and bean plants differ. Offline alternative: use a teacher-prepared printed chart of the stem sections and sketches drawn on the board if no microscopes are available.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Character change Reading Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: growth development plants primary	Learners recall the time-lapse of maize and bean seedlings. The teacher holds up a young bean seedling, a maize seedling, and a cut piece of firewood (from Eucalyptus or Cypress) showing rings. Individually learners write for 2 minutes: (a) Which parts of the seedlings are getting longer? (b) Why is the firewood stem so thick while a maize stem stays slim? (c) Where in the stem do you think new cells are made? They sketch a prediction of a cross-section of a bean stem and a maize stem, showing where growth happens. They share with a partner (Think-Pair-Share), then with the class.	Display the specimens and say: "Remember our farmer's field in Kakamega. Same day, same rain, but look at these plants. Think quietly: where does a plant get longer and where does it get thicker?" Give 2 minutes of silent think time, then say: "Turn to your partner and compare your sketches." WAIT TIME 10-15 seconds after each question before calling on anyone. Cold-call 4 pairs (not volunteers) to share, including at least 2 who predicted something different. Say: "Tell me what made you predict that." Record predictions on the board without judging. Ask: "What would we need to see to know who is right?" Clarify terms such as 'growth' and 'girth' in simple English.	Predict-Observe-Explain (POE) with Think-Pair-Share: individual prediction exposes preconceptions (for example, that plants grow all over like animals, or that all plants thicken the same way). Pairing lets learners test ideas on a peer in low-risk talk, and the class list of predictions gives a reason for the observation phase.	Observation checklist: Does the learner write a prediction with a reason? Is the sketch showing at least one location of growth (e.g. shoot tip, root tip, between bark and wood)? Listen for misconceptions: 'plants grow from the bottom', 'thickness is just the stem stretching', 'maize and beans grow the same way'. Note names of at least 6 learners with misconceptions to revisit in the Explain phase.
Observe Phase  VIDEO: Character change Reading Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: growth development plants primary	In groups of 5-6, learners rotate through 3 stations (about 8 minutes each): Station 1 – Observe a prepared slide or photomicrograph (or a hand-cut thin section stained with diluted ink or iodine) of a young dicot (bean) stem and a monocot (maize) stem under a light microscope; draw one low-power plan diagram and label epidermis, cortex, vascular bundles, xylem, phloem, and cambium (dicot only); note arrangement (ring versus scattered). Station 2 – Study a photomicrograph or chart of an older woody dicot stem showing secondary xylem, secondary phloem, vascular cambium, cork cambium, and cork, and a cut Eucalyptus/Cypress/Grevillea log	Introduce stations: "You are plant detectives. Your evidence will help us explain how maize and bean plants grow differently." Assign roles (microscopist, recorder, drawer, timekeeper) and rotate roles. Circulate and ask: "What do you see between the xylem and phloem?" "How is the maize stem different from the bean stem?" WAIT TIME 10-15 seconds after each question. Cold-call 3 groups at each station to describe one observation. Remind: "Draw what you see, not what you think should be there. Use a sharp pencil, a title, and label lines that do not cross." Safety reminder for razor blades and microscopes. Where there are not enough microscopes, point to photomicrographs or	Evidence gathering through multiple representations (microscope view, photomicrograph, real log, reading) lets learners triangulate. Using a 'draw-and-label' approach slows observation, which builds understanding of structure–function links: cambium located between xylem and phloem is the source of new cells for girth.	Observation checklist: Uses microscope correctly (focuses from low to high power) – yes/no; Drawing has title, clear lines, labels with straight pointer lines; identifies ring versus scattered bundles; notes cambium in the bean stem but not in maize; counts rings correctly. Oral probing: "Where do you think new cells are being made?" Look for evidence-based answers, e.g. 'the cambium is a thin layer between the xylem and phloem'.

	showing annual rings; count rings and record. Station 3 – Read the 'primary and secondary growth' passage and fill in a table: Where? Which meristem? What increases? Examples in maize, bean, and tree. Learners record at least three observations per station in their notebooks.	teacher-prepared charts. If no electricity, use natural light near a window for mirror-based microscopes. Prompt for simple English: "A ring of vascular bundles means..." Provide sentence starters.		
Explain Phase  VIDEO: Character change Reading Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: growth development plants primary	Learners use their drawings and notes to build a simple 3-step model of secondary growth in a woody dicot (such as Grevillea in a Kericho farm): Step 1 – young stem with vascular bundles and cambium; Step 2 – cambium forms a complete ring and produces secondary xylem inward and secondary phloem outward; Step 3 – stem thickens, cork cambium forms cork, bark develops, annual rings appear. Learners use coloured plasticine or paper strips (or draw stages on card) to make the model. Then each group writes a CER: Claim – why do bean plants thicken and branch while maize stems stay mostly the same width? Evidence – at least two observations from stations; Reasoning – linking the presence of cambium to secondary growth, apical meristems to primary growth, and monocot vascular bundles being scattered with no cambium. They also connect to the germination types: maize (hypogeal) and beans (epigeal) both start with primary growth from the shoot and root apical meristems.	Say: "Use your evidence, not just your memory. Show me where in your model the stem gets thicker." Give 5 minutes for groups to make the model, then 5 minutes for the CER. Move around and ask: "What is your evidence for that claim?" "What would happen to a Eucalyptus tree if the cambium were removed all around the stem?" WAIT TIME 10-15 seconds. Cold-call 4 groups (different from the Observe phase) to read their CER. Use the 'yes, and' technique to build on answers. Address misconceptions noticed earlier, e.g. 'plants thicken by stretching'. Give a clear teacher mini-explanation if needed: primary growth = apical meristems = length; secondary growth = lateral meristems (vascular cambium and cork cambium) = girth. Ask learners to connect: "How does this help explain our farmer's maize and beans?" Hold a Gallery Walk for 8 minutes: groups leave models and CERs on desks, visit two other groups, and leave one sticky note with 'a star' (strength) and 'a wish' (suggestion).	Claim-Evidence-Reasoning (CER) pushes learners to justify explanations with observations. Modelling secondary growth in stages makes an invisible process visible. The Gallery Walk with peer review exposes learners to alternative models and strengthens argumentation and Learning to Learn.	Peer review of models and CER using a 3-point rubric: (1) Claim is clear and answers the question; (2) at least two pieces of specific evidence cited; (3) reasoning links cambium or apical meristem to growth. Teacher listens for correct use of 'primary growth', 'secondary growth', 'cambium', 'apical meristem'. Quick check: ask 5 learners to state in one sentence the difference between primary and secondary growth. Misconceptions to flag: 'all plants have secondary growth' or 'primary growth only occurs in seedlings'.
Driving Question Board (DQB) Creation  VIDEO:	Learners return to the Driving Question Board with the main question: 'What causes the differences in germination, emergence and growth of maize	Say: "Look at the DQB. Which of our questions did today's evidence help to answer?" WAIT TIME 10-15 seconds. Invite 3 learners (cold-call) to move a sticky note to	Discussion and questioning with the Driving Question Board links each day's evidence to the unit question and builds learner ownership. Sorting questions into	Exit ticket (written on a half page): (1) 'What evidence did we gather today?' (2) 'How does this help answer our driving question?' (3) One question you still have.

Character change Reading Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: growth development plants primary	and bean plants?' In groups they write on sticky notes: (a) one thing they now know about growth, linked to evidence, and (b) one new question, e.g. 'How can we measure how fast a bean seedling grows?', 'What makes a plant stop growing?', 'Do hormones affect how stems and branches grow?'. They place the notes under categories on the DQB (Seed structure, Germination, Growth, Hormones, Measurement). The class identifies the question for the next lesson: How can you measure growth in plants?	'partly answered' or 'answered' and justify using evidence. Say: "What new questions has today raised?" Cold-call 4 groups to read new questions. Highlight the one on measuring growth: "If we want to compare maize and bean growth fairly, what would we need to measure?" If the DQB is not available, use chart paper or a chalkboard section that is preserved for the unit. Encourage learners to ask questions in simple English and help rephrase where needed.	categories shows how pieces of learning fit together and drives the next investigation.	Success indicators: names at least one specific observation (e.g. cambium between xylem and phloem, rings in a log) and links it to growth in length or thickness of maize or beans. Sort tickets into 'ready', 'nearly', and 'needs support' piles to plan Lesson 6.
Model Building Phase VIDEO: Character change Reading Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: growth development plants primary	Learners take out their Day 1 drawings and explanations about why maize and beans differ and revise them. They add a section titled 'After germination: how the plants grow', including labelled arrows showing the shoot and root apical meristems (primary growth) and lateral meristems (secondary growth in the bean and woody plants but not in maize). They use a different coloured pen to mark changes and add a short note: 'I changed ... because the evidence showed ...'. Pairs swap books and give peer feedback with one improvement suggestion. A few volunteers present revised models to the class.	Say: "Use a different colour pen so we can see how your thinking has grown. What did you add today and what evidence made you change your model?" Give 7 minutes for revision, then 3 minutes for peer review. WAIT TIME 10-15 seconds before asking pairs to share. Cold-call 3 learners to explain one change they made. Ask: "Is there anything in your model that you are not sure about yet?" Collect model checks for review and note which learners still confuse primary and secondary growth. Remind groups of the practical project: a tree nursery using stem cuttings and natural hormone sources such as onion, Aloe, and cinnamon, which will connect to growth and hormones in later lessons. Close by setting the hook: "We know plants grow in length and girth. How could we measure that? Think about how you would measure a maize seedling at home in Eldoret or Kisumu."	Model revision: learners reconstruct their explanation with new evidence, which makes learning visible and builds the habit of improving ideas with data. Using a different colour pen documents the change in thinking. Peer feedback builds scientific communication.	Model check using criteria: (1) The model includes primary growth at shoot and root tips; (2) includes secondary growth by cambium for dicots; (3) explains why maize stems differ from bean stems; (4) at least one change is justified with evidence. Quick quiz (3 questions): name the meristem that increases girth; state where primary growth occurs; give one example of a plant showing secondary growth. Learners scoring below 2 out of 3 get a short reteach in Lesson 6.

D. TEACHER REFLECTION

After the lesson, consider: Which learners could correctly distinguish primary from secondary growth using evidence, and which still hold the misconception that plants thicken by stretching? Did the microscope stations work with limited equipment, or should more photomicrographs and log sections be used next time? Were learners able to link stem anatomy to the maize and bean phenomenon? How effective was the CER writing in simple scientific English? What adjustments are needed for the next lesson on measuring growth (for example, preparing seedlings, rulers, string, and a data table for growth curves)? Did every group member participate and share equipment fairly (Unity)? Note which exit tickets show readiness to move on.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed prepared slides, photomicrographs or hand-cut sections of young bean (dicot) and maize (monocot) stems and a cut log from a woody tree with annual rings. We saw vascular bundles in a ring in the bean stem with cambium between xylem and phloem, scattered bundles in maize with no cambium, and secondary xylem, bark and rings in the older woody stem.
What did I learn?	We learned that primary growth is an increase in length produced by apical meristems at shoot and root tips, while secondary growth is an increase in girth produced by the vascular cambium and cork cambium in dicots and woody plants. Maize (a monocot) mainly shows primary growth, whereas bean and trees such as Grevillea and Eucalyptus show both. This raises the question: How can you measure growth in plants?
How does this explain the phenomenon?	Using Claim-Evidence-Reasoning, we explained that part of the difference in how maize and bean plants grow after germination is due to stem structure: the bean stem has cambium that allows thickening and branching, while the maize stem lacks cambium and grows mainly in length. This adds to our model that differences in germination and growth depend on seed structure, germination conditions, and the meristems that drive growth.

LESSON 6 (80 minutes (double lesson) or 2 x 40 minutes): Where Does a Maize or Bean Plant Grow? Measuring Growth and Regions of Growth in Seedlings

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use maize and bean seedlings to find where growth occurs in a plant, measure growth over time, and plot growth curves. This helps explain why maize and bean plants planted on the same day grow differently in height, stem thickness and root depth.
Knowledge	Growth is a permanent increase in size and dry mass. In seedlings, growth in length is concentrated in specific regions: root tips, shoot tips and, in maize, the base of the leaves. Growth can be measured as height, length, width or dry mass. A growth curve (sigmoid or S-shaped) plots growth against time and shows slow, rapid and slowing phases. This links to primary growth (apical meristems) and secondary growth (lateral meristems), which students sketched in earlier lessons.
Skills	Plan and carry out an investigation (marking seedlings with ink at equal intervals and measuring using a ruler or string); collect and record data in a table; analyse and interpret data by drawing a growth curve; construct explanations using Claim-Evidence-Reasoning; develop and revise models.
Attitudes	Unity: learners work harmoniously with peers when marking, measuring and recording data. Learning to Learn: learners reflect on their own work and adjust their measurements and explanations accordingly. Appreciation of accuracy, patience and honesty in recording data.
Key Inquiry Question	How does growth occur in plants? (Raised for next lesson: How can we measure growth, and do plants have hormones that control it?)
Purpose in Storyline	Lesson 5 showed that stems thicken and lengthen through primary and secondary growth, but it did not show WHERE growth happens on a living seedling or HOW it can be measured. Lesson 6 gives students quantitative evidence (measurements and growth curves) that growth is localised in particular regions and proceeds at different rates over time. This helps explain why the maize and bean seedlings in the farmer's field differ in height, root depth and branching. The question raised for Lesson 7 is: Do we have hormones in plants?
Safety Notes	Use washable or non-toxic ink or a pencil/thread for marking; do not use permanent markers on soft root tips. Handle seedlings gently. Wash hands after handling soil and seedlings. Dispose of seedlings and soil in the school compost or garden. Keep razor blades away from learners; the teacher alone cuts any material. Sit and move calmly around crowded benches. Use readings from the textbook (or printed page) if no seedlings are available.

B. LESSON OVERVIEW

Students return to the farmer's field: the maize and bean seeds were planted on the same day, yet the seedlings grow at different rates and in different parts. Learners predict where on a seedling growth happens, then mark young maize and bean seedling roots and shoots with equally spaced ink lines (every 5 mm or 1 cm) and observe or use prepared class data on how the marks spread apart. In groups of 4-5 they measure, tabulate and plot seedling height against time to produce a growth curve, then use the Claim-Evidence-Reasoning strategy to explain where and how growth occurs. The Driving Question Board (DQB) is updated with the new question 'Do we have hormones in plants?' and learners revise their models of maize and bean growth. The lesson relies on local materials (germinated maize and beans, ink, thread, rulers, graph paper) and needs no electricity or internet.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Growth models: introduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Exponential vs. linear growth Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the farmer's field where maize and beans were planted together after the rains and show different growth. Individually they draw a seedling of maize and of bean in their notebooks and predict: (a) Where on the seedling does growth in length take place: all along the root and shoot, or only in certain regions? (b) Will a bean seedling and a maize seedling grow at the same speed each day? They sketch a quick prediction graph of 'height against time' for 30 days. They then discuss with a partner and share with the class using Think-Pair-Share.	Display on the board a sketch of the farmer's field with maize and bean seedlings. Say: 'The farmer in Nakuru planted maize and beans on the same day. Look at your drawings from Lesson 5. Where do you think the plant gets longer?' Give 1 minute of silent individual THINK time. WAIT TIME: 10-15 seconds after asking the question before taking any answer. Then pair learners (2 minutes). Cold-call 4 pairs (not volunteers) to share, and ask: 'What makes you think that?' Record 2-3 competing predictions on the board without judging them (e.g. 'whole root grows' vs 'only the tip grows'). Clarify the words 'region' and 'growth curve' in simple English. Ask learners to hold up fingers (1-5) showing how confident they are.	Predict-Observe-Explain (POE) combined with Think-Pair-Share. Making an explicit prediction exposes prior ideas (e.g. that the whole root elongates evenly) so that the evidence in the Observe phase can confirm or challenge them. Think-Pair-Share gives every learner time to formulate and rehearse an idea before speaking, including quieter learners in large classes.	Observe and note on a checklist: Do learners draw a prediction showing a specific region of growth? Do they give a reason using Lesson 5 ideas such as 'dividing cells at the tip'? Listen for misconceptions such as 'plants grow from the base like hair' or 'growth is the same speed every day'. Count how many pairs can give a reason, not just an answer. Record the prediction on the board to return to in the Explain phase.
Observe Phase  VIDEO: The logistic growth model Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Linear vs. exponential growth: from data Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4-5, learners work with germinated maize and bean seedlings (roots 2-3 cm long, prepared by the teacher or learners from earlier lessons and grown in wet cloth, newspaper or sawdust). Step 1: Gently mark the root of each seedling with thin ink (or tie fine thread) at equal intervals of 5 mm from the tip (about 5-6 marks). Step 2: Mark the shoot (plumule/epicotyl or the stem of the bean seedling) at 1 cm intervals from the top. Step 3: Return seedlings to the moist growing medium (a plastic bag, tin or cut jerrycan with soil) and re-measure after 24 and 48 hours. Step 4: Measure the distance between the marks and record in a table. Step 5: Using a second set of seedlings grown for 4 weeks (or	Before class: prepare seedlings in advance by soaking maize and bean seeds (Kenya hybrid maize such as H614 and common beans such as Rosecoco or Mwitmania) and germinating them in damp cloth for 3-4 days. Prepare one set of seedlings per group (3 maize, 3 bean), a ruler, thread, washable ink or pencil, and graph paper. Demonstrate the marking on one seedling (touch gently and mark lightly, do not press). Circulate and ask: 'Which marks have moved apart the most? What does that tell you about where cells are dividing and growing?' and 'Why do we mark at equal distances?' WAIT TIME 10-15 seconds after each question. Cold-call at least 5 different groups to report their measurements from different	Predict-Observe-Explain (Observe step) with data tables and growth curves. Marking equal intervals makes growth visible and measurable, so learners see that growth is localised (apical regions) rather than uniform. Plotting the curve moves them from raw numbers to a pattern (slow start, rapid middle phase, levelling off), which is the pattern behind the farmer's different-sized plants.	Observation checklist during the practical: Marks made at equal intervals (Y/N); measurements recorded to the nearest mm; table complete with units; roles shared among members; seedlings handled gently. Check graphs for correct axes (time on x-axis, height on y-axis), labelled units, a smooth curve of best fit, and the three phases identified. Oral probes: 'Which interval had the fastest growth? How do you know from your graph?' Note misconceptions such as joining points with a ruler as straight lines, or claiming bean and maize curves are identical.

	class data supplied by the teacher), measure plant height every 3-4 days (day 0, 3, 7, 10, 14, 21, 28) for maize and bean and record in a table. Step 6: Plot a growth curve (height against time) on graph paper. Where seedlings are not available, learners use the data table below from a teacher-prepared handout (maize height in cm: day 0 = 1, day 7 = 4, day 14 = 14, day 21 = 32, day 28 = 48, day 35 = 55; bean: day 0 = 1, day 7 = 3, day 14 = 9, day 21 = 18, day 28 = 24, day 35 = 26) and plot both curves on the same axes. Learners note which marks moved furthest apart (root tip region) and which stayed together (older region).	regions. Prompt: 'How did the root tip differ from the older part of the root?' Ask groups to check each other's readings for accuracy (parallax, starting at zero). If connectivity is available, show the germination time-lapse video; if not, use the teacher-prepared photo set or board drawings. Remind groups of roles: marker, measurer, recorder, reporter (Unity).		
Explain Phase  VIDEO: The logistic growth model Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Exponential vs. linear growth Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups return to their predictions and use the Claim-Evidence-Reasoning (CER) strategy to explain where growth occurs and how it changes over time. Claim: 'Growth in length is concentrated at the root tip and shoot tip (apical meristems), and the rate of growth changes over time.' Evidence: they cite their own data (e.g. marks nearest the root tip moved furthest apart; maize height rose from 4 cm to 32 cm between days 7 and 21 while beans rose from 3 to 18 cm). Reasoning: they link this to cell division in meristems and cell elongation behind the tip (primary growth), to the later thickening of stems by the vascular cambium (secondary growth from Lesson 5), and to the S-shaped growth curve: slow phase (lag, seed uses stored food), rapid phase (log, active division and photosynthesis), and slowing phase (stationary, limited resources and maturity). They compare maize (monocot, fibrous	Display the CER frame: 'I claim that _____. My evidence is _____. This is because _____.' Model one sentence with a different example, e.g. 'Kenyan bamboo and sugarcane in Kisumu grow very fast at the nodes.' Ask: 'Using your data, where does growth in length happen?' WAIT TIME 10-15 seconds, then cold-call 4 groups. Press for reasoning: 'Why do the marks near the tip move apart but those higher up do not?' Introduce key terms with simple definitions on the board: meristem (region of dividing cells), elongation, primary growth (growth in length), secondary growth (growth in girth), growth curve. Ask: 'What would happen to the maize seedling if a goat nibbled the tip?' to test understanding. Link back to the phenomenon: 'How does this help explain why some farmer's plants were taller than others after two weeks?' Address misconceptions: growth is not the same as	Claim-Evidence-Reasoning (CER) with Think-Pair-Share. CER requires learners to support a claim with their own measured data and connect it to biological ideas (meristems, primary and secondary growth), which moves them from describing to explaining. Comparing statements with another group exposes gaps in reasoning and builds scientific argumentation.	Use a CER rubric: Claim is clear and specific (region named); Evidence includes at least two numerical values from the data; Reasoning links to meristems or cell division/elongation and to the three phases of the growth curve. Look for learners who use the terms 'apical meristem' and 'growth curve' correctly. Oral probing: 'What does the steep part of the curve show?' Peer review: each group exchanges CER statements and gives one 'star' (strength) and one 'step' (improvement). Note groups needing support with reasoning for re-teaching.

	roots, no secondary growth) and bean (dicot, taproot, can thicken) and suggest why the two curves differ. They then use Think-Pair-Share to compare their CER statement with another group's.	development; growth is not only height. Leave the question open: 'But what controls the cells at the tip to divide and elongate?'		
Driving Question Board (DQB) Creation  VIDEO: The logistic growth model Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Exponential vs. linear growth Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners go to the class Driving Question Board (a chart or a wall section) and revisit the main question: 'What causes the differences in germination, emergence and growth of maize and bean plants?' Each group writes on sticky notes (or paper strips): (1) what they learned today that helps answer a DQB question (e.g. 'Growth is concentrated at the tips, and a growth curve shows different rates over time'), and (2) one new question (e.g. 'Do we have hormones in plants?', 'What makes the cells at the tip divide?', 'Why does a maize stalk grow taller than a bean vine?'). They place answered questions in the 'We are beginning to answer' area and new questions in the 'New questions' area. They also complete an exit ticket individually.	Say: 'Look at the board. Which questions did we answer a little more today?' WAIT TIME 10-15 seconds. Cold-call 3 learners to place a note and justify it with evidence from their data. Guide them to the question that links to the next lesson: 'We saw that the tip grows. What do you think tells the tip to grow? Is something inside the plant sending a signal?' Highlight the question 'Do we have hormones in plants?' and circle it as the focus of Lesson 7. Hand out exit tickets: 'What evidence did we gather today? How does this help answer our driving question?' Collect for review. If the class is large, have group representatives post notes while others write exit tickets.	Discussion and questioning using the Driving Question Board. The DQB links each lesson to the anchoring phenomenon, helps learners see their progress, and turns gaps in understanding into the next question, creating a coherent storyline from measurement to hormones.	Exit ticket (collect and review): indicators of success are (a) names at least one piece of evidence (e.g. 'marks near the root tip moved apart 8 mm') and (b) links it to the driving question (e.g. 'different parts and stages of growth help explain why the plants differ'). Check that the new question posted is productive (e.g. about what controls growth). Sort tickets into three piles (secure, partly secure, needs help) to plan Lesson 7 groupings.
Model Building Phase  VIDEO: The logistic growth model Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Exponential vs. linear growth Source: Khan Academy (English - US curriculum)	Learners revisit their Day 1 drawing and explanation of how maize and bean plants grow and revise their models. Each group updates a poster-size model (on manila paper) showing: a maize and a bean seedling with labelled regions of primary growth (apical meristems at root tip and shoot tip, region of elongation) and secondary growth (vascular cambium thickening in the bean stem); a small growth curve with the three phases labelled; and arrows showing factors (water, oxygen, temperature, stored food, light) that influence growth. They	Hand out Day 1 drawings from the learner folders. Say: 'Compare your first idea with what you know now. What would you change and why?' WAIT TIME 10-15 seconds. Provide model criteria on the board: labelled regions of growth; growth curve with phases; link to the farmer's maize and beans; evidence used. During the Gallery Walk (6-8 minutes), rotate groups every 2 minutes and cold-call 3 groups to share the best suggestion received and how they will use it. Ask: 'What can your model explain now that it could not explain on Day 1? What can it not	Model revision plus Gallery Walk with peer review. Revising the Day 1 model with new evidence shows learners how their thinking has changed, and peer comments strengthen explanations. Leaving a 'missing piece' in the model (what controls growth) builds motivation for Lesson 7 on plant hormones.	Model check using a rubric: regions of growth correctly labelled (apical meristems, elongation zone); distinguishes primary from secondary growth; growth curve included with three phases; link to maize and bean differences; evidence cited; at least one peer comment received and acted upon. Observe whether learners add accurate improvements, not just decoration. Quick quiz (3 questions) if time allows: (1) Name the region where root growth is greatest. (2) What is shown by the steep part of an S-shaped curve? (3) Which plant, maize or bean,

Search ARES for similar readings	add a 'What is still missing?' box with a question mark for what controls the tip (to be explored in Lesson 7). Groups then do a short Gallery Walk: they display models, move to another group's poster, and leave one comment using 'I like... I wonder... A suggestion...'.	yet explain?' Reinforce the need for an explanation of control (hormones). Display models in the class for use in later lessons. Remind learners that the tree nursery project later in the unit will need them to understand where roots and shoots grow on cuttings.	undergoes secondary growth? Record those who need further support.
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D. TEACHER REFLECTION

After the lesson, reflect: Did all groups mark and measure accurately, and did the data show clear localised growth? Which learners could write a full CER statement with numerical evidence, and who relied only on description? Were there misconceptions (e.g. whole root grows evenly, growth equals height only, growth curve is a straight line)? Did sharing of rulers and seedlings work in a large class, and would pre-marked seedlings or the teacher-prepared data table have helped? Was WAIT TIME used consistently and did cold-calling include quieter learners? Did the DQB question 'Do we have hormones in plants?' arise from learners themselves? Plan adjustments for Lesson 7 using the exit tickets, for example by grouping learners who need support with growth curves.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We marked maize and bean seedlings at equal intervals and measured how the marks moved apart over time. We also measured seedling height at intervals and plotted a growth curve. The marks near the root tip and shoot tip moved apart the most, and the curve showed a slow start, a rapid middle phase, and a levelling-off phase.
What did I learn?	Growth in length is concentrated in specific regions (apical meristems at the root and shoot tips, followed by elongation), while stems thicken through secondary growth in some plants such as beans. Growth can be measured and shown on a growth curve that is S-shaped. Maize and bean seedlings grow at different rates and in different ways.
How does this explain the phenomenon?	The maize and bean plants planted on the same day differ in height, root depth and branching because growth occurs in particular regions and at different rates in each plant type and over time. The sigmoid growth curve and the types of growth (primary and secondary) help explain these differences. What controls the cells at the tips to divide and elongate is still unknown, which raises the next question: Do we have hormones in plants?

LESSON 7 (80 minutes (double period)): Plant Hormones: What Controls Growth in Maize and Bean Seedlings?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use a video and a reading on plant hormones to work out how chemical messengers coordinate germination, shoot and root growth, branching and dormancy. They then apply this to explain why maize and bean seedlings planted on the same day with the same rain grow differently. The lesson also links back to the primary and secondary growth modelled earlier and leads to the question: how can we measure growth?
Knowledge	Learners will be able to demonstrate the role of hormones in plant growth and development: auxins (cell elongation, apical dominance, rooting of cuttings), gibberellins (breaking dormancy, stem elongation, germination), cytokinins (cell division, lateral bud growth), abscisic acid (dormancy, closing stomata) and ethylene (fruit ripening).
Skills	Developing and using models; analysing and interpreting data; constructing explanations using Claim-Evidence-Reasoning; obtaining, evaluating and communicating information from a video and a reading; asking questions.
Attitudes	Unity: works harmoniously with others while discussing evidence in groups. Learning to Learn: reflects on own work and adjusts it as new evidence appears. Appreciation of the importance of growth and development in plants (SLO h). Financial Literacy and Prevention of Global Warming: links hormones to tree nursery cuttings and seedling sales.
Key Inquiry Question	How does growth occur in plants?
Purpose in Storyline	Lessons 1-6 covered seed structure, dormancy, germination conditions and types, and primary and secondary growth. This lesson adds the chemical control behind those processes: why seeds that look healthy differ in germination time, and why stems elongate, roots deepen and branches form. It creates a need to measure growth (the question raised for Lesson 8) and connects to the tree nursery project, where natural hormone sources such as bulb onions, Aloe and cinnamon help roots form on cuttings.
Safety Notes	No hazardous chemicals are used. If learners handle seedlings or soil, they should wash hands afterwards. Do not taste any plant material or seeds, as some bean seeds are treated with pesticide. Keep walkways clear during the Gallery Walk. Natural rooting materials (onion, Aloe, cinnamon) are for external use only, and anyone with skin allergies should avoid them.

B. LESSON OVERVIEW

The lesson opens with the anchoring phenomenon: maize and beans planted together after rain behave differently. Learners predict what invisible internal factor could explain this. They watch the Khan Academy video on major plant hormones (or hear the teacher-prepared narration if there is no connectivity) and read the accompanying notes on hormone roles. They record evidence in a hormone table and use Think-Pair-Share and CER to explain parts of the phenomenon. The Driving Question Board (DQB) is updated, and learners revise their Day 1 model by adding hormones as the control system. The lesson ends with an exit ticket and the new question: how can we measure growth?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners look at the phenomenon	Display the sketch and say: 'Same	Think-Pair-Share, within Predict-	Observation checklist: (1) each

 VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: growth development plants hormones  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	sketch on the board (time-lapse stills or a teacher drawing): maize and bean seeds planted the same day in a shamba in Kakamega after rain. Some seedlings are tall, some are short, one bean has side branches and one seed has not germinated. On a half-page they individually write a prediction for 'What inside the plant could be controlling when a seed germinates, how tall the shoot grows and whether branches form?' and draw a quick arrow diagram. They then share with a partner (Think-Pair-Share) and add one idea from the partner in a different colour.	day, same rain, same shamba. Why do these plants behave differently?' Give 10-15 seconds of silent WAIT TIME before anyone speaks. Say: 'Write one idea. Think about what messages a plant might use, since it has no brain or nerves.' Allow 3 minutes for writing, then 3 minutes for pairs. Cold-call 4 pairs (not volunteers) with: 'Tell us your partner's idea, not your own.' Record words such as 'chemical', 'food store' and 'water' on the board without judging. Circle any learner idea about 'chemical messengers' and say: 'Keep that word in mind; we will test it today.'	Observe-Explain (POE). Individual thinking lets every learner commit to an idea before the video. Pair talk makes learners voice and refine reasoning. Saying a partner's idea builds listening and unity. Recorded predictions give a baseline to compare with video evidence later.	learner writes a prediction and a diagram within 3 minutes; (2) at least 3 of 4 cold-called learners accurately restate their partner's idea; (3) count learners who mention a chemical or internal signal rather than only external factors such as water or soil. Oral probe: 'What makes you think that?' Note misconceptions such as 'plants grow only because of rain' or 'plants cannot send signals'.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: growth development plants hormones  EXERCISE: Plant Hormones Practice Source: CK-12  Search ARES for similar readings	In groups of 4-5 (sharing one textbook or reading per group), learners watch the Khan Academy video 'Major plant hormones (& how to remember) Control & Coordination class 10 Biology' (or hear the teacher's narrated version with chalkboard drawings if there is no electricity or internet). Each group fills in a hormone evidence table with columns: Hormone Where made What it does Link to maize/bean phenomenon. Roles within the group: Reader, Recorder, Reporter, Materials/Time keeper. After the video, groups read the notes on the role of plant hormones and their functions and add details to the table. They highlight at least one example for each of: auxin (shoot elongation, apical dominance, rooting of cuttings), gibberellins (breaking dormancy, germination, stem elongation), cytokinins (cell division, lateral bud growth),	Before the video say: 'Watch for one hormone that could explain a seed that is healthy but has not germinated, and one that could explain side branches.' Pause the video at each hormone and ask: 'What did you see or hear?' Give 10-15 seconds WAIT TIME, then cold-call 2 groups per pause. Offline fallback: read the narration aloud in short chunks and draw the hormone effects on the board (a tall maize shoot, a bean with side branches, a ripening banana). Clarify terms: 'elongation means getting longer', 'dormancy means a resting seed', 'lateral bud means a side bud'. Circulate and ask: 'Which part of the table helps explain the seed that did not germinate?' Remind groups to share books and to write in simple, clear scientific English. Link to the tree nursery project: 'Why might bulb onion or Aloe extract help cuttings form roots?' (natural rooting hormones such as auxins).	Predict-Observe-Explain (Observe step) with structured evidence recording. The table makes learners pull evidence from two sources (video and reading) and tie each hormone to a part of the phenomenon. Pausing the video for prediction and discussion keeps learners actively connecting new information to their earlier predictions.	Observation checklist during group work: (1) every group has at least 4 hormones recorded with correct location and function; (2) each learner fulfils a role; (3) the 'Link to phenomenon' column is filled with a specific link (e.g. 'gibberellins could break dormancy in the bean that has not germinated'). Oral probes: 'Which hormone makes stems longer?' and 'What does apical dominance mean?' Watch for the misconception that hormones are only found in animals or that one hormone does only one job.

	abscisic acid (dormancy, stomatal closure) and ethylene (fruit ripening, e.g. bananas or mangoes sold in Mombasa).			
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: growth development plants hormones  HTML: Plant Hormones - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Using the evidence table, groups write a Claim-Evidence-Reasoning (CER) explanation for one part of the phenomenon assigned by the teacher: (A) a healthy bean seed that germinates late or not at all; (B) maize shoots that elongate quickly; (C) bean plants that develop side branches while maize stays unbranched; (D) farmers in Kericho or Nyeri rooting tea or tree cuttings with natural hormone sources. Each group writes: Claim (what hormone/process explains it), Evidence (from video, reading or earlier lessons such as seed dormancy and germination conditions), Reasoning (how the hormone causes the effect, linking to cell division, elongation or secondary growth). Groups then do a short Gallery Walk, reading other CERs and leaving one sticky note or pencil comment ('I agree because...' or 'What about...?').	Model one CER on the board, e.g. 'Claim: Auxin from the tip of the bean stem keeps side buds from growing. Evidence: the video showed that removing the tip allowed side shoots to grow. Reasoning: auxin moves down and suppresses lateral buds, so removing the tip lowers auxin and lets cytokinins promote branching.' Say: 'Your reasoning must say how, not just what.' Allow 8 minutes writing. Say: 'Stop and look at your evidence. Does it support your claim?' Give 10-15 seconds WAIT TIME. For the Gallery Walk, set 6 minutes and rotate groups every 2 minutes. Cold-call 3 groups to share one comment they received and how they will revise their explanation. Probe: 'Which factor besides hormones also affects germination, such as water, oxygen or temperature? How do they interact?' Clarify that hormones respond to environmental conditions and do not replace them.	Claim-Evidence-Reasoning (CER) plus Gallery Walk. CER forces learners to justify explanations with evidence and causal reasoning, and the Gallery Walk exposes them to different parts of the phenomenon and to peer critique, helping them build a connected explanation of why maize and beans differ.	Use a CER rubric (0-2 per part): the claim names a correct hormone or process; the evidence is drawn from the video, reading or earlier lessons; the reasoning explains the mechanism. Success indicator: at least 80% of groups reach a score of 4 or more out of 6. Peer review: each group gives one specific comment. Oral probes identify misconceptions such as 'hormones are food for the plant' or 'gibberellins cause dormancy'.
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: growth development	Learners revisit the class DQB (on manila paper or the chalkboard) holding the driving question, 'What causes the differences in germination, emergence and growth of maize and bean plants?' In pairs they write on sticky notes or paper strips: (1) one thing they now understand about hormones and growth, and (2) one new question. They place them on the DQB under the headings 'Answered', 'Partly answered' and	Say: 'Look at our board. What can we now move from questions to answers?' Allow 10-15 seconds WAIT TIME. Invite 3 pairs (cold-called) to move a question and justify with evidence from today's video or reading. Say: 'Which new question do we need data to answer?' Guide learners to the point that to compare maize and bean growth, we need measurements, introducing Lesson 8. Tag questions by type	Discussion and questioning with the Driving Question Board. Moving questions across categories shows learners how evidence advances understanding. Generating new questions keeps them engaged in inquiry and links each lesson to the driving question.	Check that at least 80% of pairs write one accurate statement using the term 'hormone' and one testable question. Observe whether learners justify the placement of a note with evidence. Look for questions that move from 'what' to 'how' or 'why'. Note any unresolved confusions to address in Lesson 8.

plants hormones  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	'New questions'. They add the new question 'How can you measure growth in plants?' and others such as 'Do maize and beans have the same amounts of hormones?', 'Can farmers use hormones to improve yields?' and 'How can we show growth on a graph?'	(can investigate with seedlings, need reading, need video). If paper is scarce, use the chalkboard and one scribe per group.		
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: growth development plants hormones  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners retrieve their Day 1 model/drawing of what causes differences in maize and bean germination and growth. Individually, in a different colour, they revise it to include: (a) hormones (gibberellins, auxins, cytokinins, abscisic acid) with arrows showing where they act on the seed, shoot, root and lateral buds; (b) links to earlier lessons (dormancy, germination conditions, primary growth at root and shoot tips, secondary growth in woody dicots, e.g. a mukau or jacaranda tree); and (c) a short caption explaining one cause of the differences in maize and beans. They swap with a partner for peer review using two stars and a wish.	Say: 'Your Day 1 model was your first guess. What would you add or change now, and what evidence made you change it?' Give 10-15 seconds WAIT TIME before they start. Circulate, and ask 5 learners: 'Where is the hormone made, and what does it do here?' Prompt: 'Show what happens when the shoot tip is cut.' Remind learners to label clearly. After peer review, cold-call 3 learners to share one revision and its evidence. Collect the models for a model check. Hand out exit tickets: 'What evidence did we gather today?' and 'How does this help answer our driving question?'	Model revision. Learners compare earlier ideas with new evidence, which strengthens conceptual change and shows learning growth (Learning to Learn). Peer review gives feedback on clarity and accuracy.	Model check: look for at least three hormones with correct locations and functions, at least one link to dormancy or germination, and one link to branching or elongation. Peer review: two stars and a wish with specific wording. Exit ticket: success if the learner names one piece of evidence (e.g. 'the video showed that removing the shoot tip allowed side shoots to grow') and links it to the driving question. A quick quiz of 3 questions (name the hormone for dormancy-breaking, elongation, ripening) can be used for the last 3 minutes.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners connect hormone roles to the maize and bean phenomenon, or only list functions? (2) Which hormone ideas caused confusion (e.g. gibberellins vs auxins, or apical dominance)? (3) Did group roles work with shared textbooks and large classes? (4) Was the offline narration as effective as the video? (5) How did predictions at the start compare with final models? Use exit tickets to sort learners into those who need re-teaching on hormone functions, those ready for measuring growth in Lesson 8, and those who could extend to agricultural uses such as rooting powders in tree nurseries. Plan to bring maize and bean seedlings (or marked seedlings grown in tins) next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We watched a video and read about the major plant hormones (auxins, gibberellins, cytokinins, abscisic acid and ethylene) and
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	recorded where each is made and what it does in the plant.
What did I learn?	Hormones are chemical messengers that coordinate germination, dormancy, shoot and root growth, branching and fruit ripening. They work together with environmental factors such as water, oxygen and temperature.
How does this explain the phenomenon?	Differences in germination, emergence and growth between maize and bean plants can partly be explained by hormone balance: for example, gibberellins and abscisic acid affect dormancy and germination time, auxin affects shoot elongation and apical dominance, and cytokinins affect branching. To compare growth we now need to measure it.

LESSON 8 (80 minutes (double period), with daily 5-minute measurement checks beforehand): Measuring Growth: Seedling Height, Stem Width and Growth Curves of Maize and Beans

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners record seedling height and stem width data from maize and bean seedlings, plot growth curves, and use the curves to explain why the two crops, planted on the same day after the same rain, grow at different rates. The lesson ends with the new question: Do we have hormones in plants?
Knowledge	Growth is a permanent increase in size, dry mass or cell number. It can be measured as height, stem width, leaf area, fresh mass or dry mass. A typical growth curve is S-shaped (sigmoid), with a lag phase, a rapid (exponential) phase and a stationary phase. Maize (monocot, epigeal/hypogeal context from earlier lessons) and beans (dicot) have different growth patterns and rates.
Skills	Measuring with a ruler, string and thread. Recording data in a table. Plotting line graphs. Interpreting graph shapes. Making Claims with Evidence and Reasoning (CER). Working in groups and giving peer feedback. (NGSS SEP: Analyzing and Interpreting Data; Using Mathematics and Computational Thinking; Engaging in Argument from Evidence.)
Attitudes	Unity: works harmoniously with group members while measuring and recording. Learning to Learn: reflects on own data and graphs and corrects errors. Appreciation of the importance of growth and development in plants. Curiosity about what controls growth.
Key Inquiry Question	How does growth occur in plants? Sub-question for today: How can we measure growth, and what do growth curves tell us about why maize and beans grow differently?
Purpose in Storyline	Lessons 1-7 gave evidence on seed structure, dormancy, germination conditions and regions of growth. Lesson 8 turns observations into quantitative data. The curves show that growth is not constant: it speeds up and slows down, and maize and beans differ. This raises the question of what controls the timing and speed of growth, leading to Lesson 9 on plant hormones (Do we have hormones in plants?).
Safety Notes	Handle seedlings gently and wash hands after touching soil. Use rulers and thread, not sharp blades. Keep walkways clear of tins and trays. Wash hands before leaving the class. No electricity or internet is needed; the time-lapse video is optional, and the teacher's chart is the alternative.

B. LESSON OVERVIEW

Learners return to the farmer's puzzle: maize and beans planted on the same day after the same rain look different as they grow. Using seedlings grown in tins, cartons or sacks since Lesson 5 (or teacher-prepared data sets for days missed), groups measure shoot height with a ruler and stem width with thread wrapped around the stem and measured against the ruler. They compile data for Days 0-21, plot growth curves on graph paper, compare the S-shaped curve of one crop against the other, and write CER explanations. A Gallery Walk lets groups review each other's graphs. The DQB gains the question 'Do we have hormones in plants?' and the class revises its model of growth.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Logistic growth versus exponential growth Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the farmer's field with maize and beans. Teacher draws empty axes on the board (height in cm vs time in days). Individually, learners sketch on a half-page what they predict the height of a maize seedling and a bean seedling will look like over 21 days: a straight line, a curve that keeps rising, or something else. They also predict which will be taller at Day 7 and Day 21, and write one reason. They then pair up and compare, then share with the class (Think-Pair-Share).	Say: 'Imagine you walk past the farmer's shamba every day for three weeks with a ruler. Sketch what the height of the maize and beans would look like on a graph. Do not talk yet.' Give WAIT TIME of 10-15 seconds, then 2 minutes of silent sketching. Say: 'Turn to your partner. Show your sketch and explain why you drew it that way.' Circulate and listen for 'it grows the same amount every day'. Cold-call 4 pairs to describe their sketches; draw 2-3 different sketches on the board without judging them. Ask: 'Does anyone think the growth rate changes? Why?' WAIT TIME 10 seconds.	Predict-Observe-Explain (POE) with Think-Pair-Share. Predictions make learners commit to an idea about the shape of growth, such as a straight line, so the real data can challenge it. Pair talk lets every learner voice reasoning before the whole class hears it.	Observation indicators: learners draw a labelled axis with units; give a reason linked to the phenomenon (for example 'maize is a grass so it grows tall quickly'); at least 4 pairs share. Note the number of learners who predict a straight line, which is a misconception to revisit. Record the count on the checklist.
Observe Phase  VIDEO: Exponential vs. linear growth Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Stem Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of 4-5 with roles (measurer, reader, recorder, checker), learners measure the height of 3 maize and 3 bean seedlings from the class nursery, from soil level to the shoot tip, in cm. They measure stem width by wrapping thread around the stem once, then measuring the thread length against a ruler and dividing by 3 (circumference / 3 $\approx$ diameter). They add today's readings to the group table from earlier days (Days 0, 3, 6, 9, 12, 15, 18, 21). Where seedlings were missed, the teacher gives a back-up data table (for example maize mean height: Day 3 = 2 cm, 6 = 8, 9 = 18, 12 = 32, 15 = 48, 18 = 58, 21 = 62; bean mean height: Day 3 = 1, 6 = 3, 9 = 7, 12 = 12, 15 = 17, 18 = 20, 21 = 21). Learners calculate mean values for each day, plot both height curves on graph paper using different colours, and plot stem width as a second graph if time permits. They note where the curve is flat, steep	Demonstrate once: 'Place the zero of the ruler at the soil surface, not at the pot edge. Read at eye level.' Say: 'Measure three seedlings per crop, then find the mean. Why do we use more than one seedling?' WAIT TIME 10 seconds, then cold-call 2 learners. Circulate with the observation checklist. Prompt: 'Which part of your curve is steepest? Which days are slow?' If groups plot a straight line through points, ask: 'Do all your points lie on your line? What does the curve actually do?' Pre-print or draw a graph template on the board for groups with limited graph paper. Label the phases: lag, rapid, stationary. Without electricity, use a teacher-prepared chart in place of the time-lapse video; if power is available, show 2 minutes of the time-lapse after plotting.	Data collection with Analyzing and Interpreting Data, and visual inference. Turning numbers into graphs lets learners see patterns of changing growth rates that are hidden in the table, building a quantitative picture of the farmer's puzzle.	Observation checklist: ruler zero at soil level; reading at eye level; thread method used correctly; mean calculated correctly; axes labelled with units; points plotted accurately (within half a small square); roles shared fairly. Oral probe: 'Why is the bean curve lower than the maize curve at Day 15?' Note learners who confuse growth rate (slope) with total height.

	or levelling off.			
Explain Phase  VIDEO: Per capita population growth and exponential growth Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Exponential Growth (Flexbook) Source: CK-12  Search ARES for similar readings	Groups interpret their curves: they identify the lag phase (slow start while the seed absorbs water and mobilises food stores), the exponential phase (rapid cell division and elongation), and the stationary phase (growth slows as the plant reaches maturity or resources limit it). They write a CER statement: Claim (for example 'Maize grows faster than beans between Days 6 and 15'), Evidence (specific numbers from their table and graph), Reasoning (links to cotyledon food stores, shoot structure, leaf area for photosynthesis, and the monocot/dicot differences from Lessons 1-7). They also compare height with stem width: maize gains height quickly, while bean stems thicken differently. Pairs swap CER statements and give one star (strength) and one wish (improvement). A short whole-class discussion links to the farmer's puzzle and to the seeds that failed to germinate (no data, so no curve).	Say: 'Point to the steepest part of the maize curve. What was happening in the seedling then?' WAIT TIME 10-15 seconds. Cold-call 3 groups. Provide the CER frame on the board: 'I claim... The evidence is... This is because...'. Press for numbers: 'Which two values show that? Subtract them for me.' Clarify difficult terms: sigmoid means S-shaped; exponential means increasing faster and faster. Ask: 'Planted on the same day with the same rain, why do the curves still differ?' Accept ideas such as seed size, food reserves, genes, temperature, soil nutrients. Then ask: 'Something inside the plant seems to speed up and slow down growth at different times. What could be controlling that?' WAIT TIME 15 seconds. Do not give the answer; let it lead to the DQB.	Claim-Evidence-Reasoning (CER) with peer review. CER forces learners to tie each statement about growth to specific numbers and biological reasons instead of opinion, and star/wish feedback develops scientific argumentation.	Rubric for CER (0-3): claim is testable; evidence cites at least two numbers from the graph; reasoning links to biology (food stores, cell division, leaf area). Indicator of mastery: learner correctly names the three phases on the curve. Quick oral quiz of 3 questions: 'What does the steep part of the curve show?' 'Why does the curve level off?' 'Which measurement tells us about thickening?' Note misconceptions such as 'a flat line means the plant has died'.
Driving Question Board (DQB) Creation  VIDEO: Exponential growth & decay word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Exponential Growth Practice	Learners return to the class DQB. In groups they write on sticky notes or paper: (1) one new piece of evidence from today (for example 'Maize shows a rapid growth phase between Days 6 and 15') and (2) one new question. They place these under the matching DQB categories (seed structure, dormancy, germination conditions, growth measurement). The class notices that many questions ask what controls the speed and timing of growth and agrees on the lesson's new question: 'Do we have hormones	Say: 'Look at our Driving Question: What causes the differences in germination, emergence and growth of maize and bean plants? What does today's graph add?' WAIT TIME 10 seconds. Cold-call 3 groups to post notes and read them aloud. Read the question aloud and say: 'Animals have hormones such as insulin and adrenaline. Do we have hormones in plants? What evidence would we need?' WAIT TIME 15 seconds. Write the question in a coloured marker at the top of the DQB for Lesson 9. Ask: 'Which	Discussion and questioning using the Driving Question Board. Learners link today's data to the driving question and see their own questions shaping the next lesson, which builds ownership and coherence.	Indicators: sticky notes include evidence rather than only opinions; new questions are testable; at least 5 groups contribute; learners move or tick at least one earlier question. Note whether learners can explain why the hormone question follows from the curves. Exit ticket: 'What evidence did we gather today?' and 'How does this help answer our driving question?'

Source: CK-12 Search ARES for similar readings	in plants?' Learners also move at least one earlier question to 'partly answered'.	earlier questions can we now partly answer?' and have learners tick them. If no wall space is available, use manila paper or a chalkboard section.		
Model Building Phase  VIDEO: Exponential vs. linear growth over time Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Exponential Growth and Decay (Flexbook) Source: CK-12 Search ARES for similar readings	Learners revisit their Day 1 drawings and explanations of why maize and beans grow differently. Individually, they add or revise parts of the model: growth curve sketches for maize and beans with the three phases labelled; arrows showing food stores (endosperm in maize, cotyledons in beans) feeding early growth; and a question mark labelled 'something controls the speed? (hormones?)'. Groups then take part in a Gallery Walk, posting their graphs and revised models; each learner visits two other groups, leaves one comment on a sticky note ('I agree because...' or 'Have you considered...'), and returns to improve their own model using the feedback. Early finishers add a note on how a farmer in Kericho or Kitale could use growth curves to decide when to top-dress maize or weed beans.	Say: 'Take out your Day 1 drawing. Use a different colour pen to show what you now know.' WAIT TIME 10 seconds, then 5 minutes of silent revision. Set the Gallery Walk: 'You have 8 minutes. Visit two groups. Leave one comment that refers to evidence.' Circulate and ask: 'Where on your model is the evidence from today's graph?' Cold-call 3 learners to describe one change they made and why. Close with the exit ticket on paper or as an oral round of 5 learners. Assign homework: observe and record a plant at home (for example sukuma wiki or a tree seedling) and bring one question about what makes it bend toward light.	Model revision with Gallery Walk and peer review. Updating the Day 1 model with measured evidence shows learners how their understanding has changed. Peer comments push them to justify the model with data, which prepares them to add hormones in Lesson 9.	Model check indicators: revised model includes the three growth phases; food store arrows; differences between maize and beans; and an unanswered question about control. Peer review comments cite evidence. Exit ticket sorting: 'Fully', 'Partly' or 'Not yet' for the target 'I can draw and interpret a growth curve'. Follow up with learners who place the stationary phase incorrectly.

D. TEACHER REFLECTION

Which groups showed accurate measuring and plotting, and which struggled with scale or the thread method? Did learners move beyond 'maize is taller' to explain the change in growth rate using phases of the curve? How many learners still predicted straight-line growth after the data? Which Kenyan contexts (the farmer's shamba, tins of maize and beans, top-dressing in Kitale or Kericho) helped learners connect to the phenomenon? Was there enough time for the Gallery Walk without electricity or full graph paper? What adjustments are needed for large classes (for example, one measurement station per group and shared rulers)? Which learners need support with terminology (sigmoid, exponential, stationary) before Lesson 9 on plant hormones?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We measured the height and stem width of maize and bean seedlings over time and plotted growth curves. Both curves were S-shaped, with a slow start, a rapid growth phase and a levelling-off phase, but maize grew faster and taller than beans in the same
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	period.
What did I learn?	Growth can be measured as height, stem width, mass or leaf area. A growth curve shows that growth rate changes over time (lag, exponential and stationary phases). Maize and beans differ in growth rate and pattern even when planted on the same day.
How does this explain the phenomenon?	The growth rate changes because of food stores, cell division and elongation, leaf area for photosynthesis and the type of plant. Something inside the plant may be controlling the timing and speed of growth, so we ask: Do we have hormones in plants?

LESSON 9 (80 minutes (double lesson)): Role of Hormones in Plant Growth and Development: Why Do Maize and Bean Plants Grow Differently?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate how plant hormones coordinate germination, shoot and root growth, and branching, and use this evidence to explain why maize and bean plants planted on the same day after the same rain differ in germination, emergence and growth.
Knowledge	By the end of the lesson, the learner should be able to: demonstrate the role of hormones in plant growth and development (Specific Learning Outcome g); name the major plant hormones (auxins, gibberellins, cytokinins, abscisic acid, ethylene) and state one function of each; explain apical dominance and how removing the shoot tip affects branching; relate hormone action to agriculture, e.g. pruning of tea in Kericho, rooting of cuttings, ripening of mangoes and bananas.
Skills	Developing and using models (hormone flow diagram); analysing and interpreting data from a video and a reading; constructing explanations using Claim-Evidence-Reasoning; obtaining, evaluating and communicating information; recording data and drawing conclusions from a simple pruning demonstration.
Attitudes	Unity: works harmoniously with peers when analysing evidence and building a shared model. Learning to Learn: reflects on and revises own Day 1 ideas. Citizenship and Patriotism: values sharing seedlings and farming knowledge with the community. Financial Literacy: appreciates how hormone knowledge (pruning, rooting cuttings) can raise farm and nursery income. Prevention of Global Warming: appreciates tree nurseries as a way to mitigate climate change.
Key Inquiry Question	How does growth occur in plants? (Lesson focus: What chemical messages inside the plant control when seeds germinate and how seedlings grow and branch?)
Purpose in Storyline	Lessons 1-8 gathered evidence on seed structure, dormancy, germination conditions and types, primary and secondary growth, and growth curves. Learners now have a gap: the structures and conditions alone do not explain why seeds under identical rain behave differently or why some plants branch while others stay single-stemmed. Lesson 9 supplies the internal control mechanism, hormones, and sets up Lesson 10, the final revision of the class model and the claim on the driving question.
Safety Notes	Pruning demonstration: use clean, sharp scissors or a razor blade handled only by the teacher or one trained learner per group; cut away from the body. Do not taste any plant materials (onion, Aloe, cinnamon, bean seedlings) used in the demonstration. Wash hands after handling soil and plants. Keep seedling trays away from walkways. If any learner is allergic to Aloe sap or onion, they act as recorder only. Do not consume treated seeds or any seedlings from the experiment.

B. LESSON OVERVIEW

Learners begin by predicting what would happen if the tip of a young bean seedling were cut off, using a Think-Pair-Share. They then gather evidence from a time-lapse or teacher-prepared picture sequence (offline alternative: large chart drawings) showing hormone effects, and read a short passage on auxins, gibberellins, cytokinins, abscisic acid and ethylene, including apical dominance. A pruned versus unpruned bean seedling demonstration (set up earlier or shown live) supports the evidence. In the Explain phase learners write a Claim-Evidence-Reasoning statement on how hormones help explain differences in germination, emergence and growth of maize and beans, and connect it to Kenyan agriculture (tea plucking in Kericho, rooting cuttings with onion or Aloe in a school tree nursery, ripening bananas in Mombasa markets). They add questions and answers to the Driving Question Board and revise their Day 1 model by adding hormone arrows and labels. The lesson uses small groups, shared textbooks and locally available materials, and does not depend on electricity or the internet.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Phototropism Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the farmer's maize and bean field after rainfall. The teacher shows a bean seedling in a cup (or a labelled drawing on the board) and asks: 'If a farmer pinches off the very top of this young bean plant, what will happen to the stems over the next two weeks?' Learners think silently for 1 minute and write a prediction and a reason in their notebooks. They pair up, compare predictions, then share. They also predict: 'What might be moving inside the plant that tells the seed when to germinate and the shoot when to grow?' Predictions are recorded on a class chart titled 'What we think controls plant growth'.	Say: 'Remember our farmer's field. Same rain, same day, but different results. Today we ask what is happening inside the plant.' Hold up the seedling and say: 'Write your prediction in one sentence and give a reason beginning with "because".' Give 60 seconds of silent think time. Then say: 'Turn to your partner and compare.' Circulate for 2 minutes and listen for ideas such as 'the plant will grow sideways' or 'the plant will die'. Ask: 'What do you think is sending signals in the plant?' WAIT TIME 10-15 seconds before taking answers. Cold-call 4 learners from different groups, including at least 2 who have not spoken yet, and record each prediction without judging it. Ask one follow-up: 'Who can tell us why they chose that?'	Think-Pair-Share and Predict-Observe-Explain (POE): Individual thinking surfaces each learner's own ideas before any instruction, pairing lets learners rehearse and refine their reasoning in a low-risk setting, and public sharing creates a class record of predictions to test against evidence later. The prediction is directly linked to the phenomenon of different growth patterns in the same field.	Observation: Tick on a checklist how many learners write a prediction with a reason (target: at least 80%). Look for the misconception that plants grow only because of water and soil, with no internal control. Note learners who use words such as 'signal', 'chemical' or 'tip'. Oral probe: 'What makes you think the tip matters?' Learners who cannot give a reason are flagged for support during the Observe phase.
Observe Phase  VIDEO: The logistic growth model Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of 5-6, learners gather evidence from three sources. (1) Video or picture sequence: they watch a short time-lapse of seedlings responding to hormones (if available; offline alternative: the teacher shows a prepared chart of four drawings: a seedling with intact tip, a seedling with the tip removed, a tip with an agar or soil block placed back on it, and a seedling bent toward light), and complete an observation table of what changed and where. (2) Reading: each group receives one section of the 'Plant Hormones' reading card (auxins, gibberellins, cytokinins, abscisic acid and ethylene, plus apical dominance), and records the function and a	Say: 'You are now scientists collecting evidence. Write only what you can see, not what you think.' Before the video or chart, say: 'Watch for three things: where the change happens, what part of the plant responds, and how long it takes.' Play the video twice if available; if there is no electricity or internet, say: 'Our video is on paper today. Look closely at each drawing.' and walk the chart through step by step. WAIT TIME 10-15 seconds after asking: 'What is different between the seedling with the tip and the one without?' Cold-call 3 groups to report one observation each. Distribute reading cards (one per group, shared between 5-6 learners) and	Predict-Observe-Explain (POE), Observe step: learners test their earlier predictions against three independent sources of evidence (video or chart, text and a live demonstration). Triangulating sources helps them see that the same pattern appears repeatedly, which strengthens their later claims. Jigsaw-style sharing of reading cards also spreads the reading load, which suits large classes with limited textbooks.	Observation checklist: tick learners who (a) record observations without interpreting, (b) measure height correctly in cm, (c) count side branches accurately, and (d) take a role in the group. Success indicators: the observation table shows that the pruned seedling has more side branches and the intact seedling is taller; each group's reading table has the correct function for its hormone (e.g. auxin: cell elongation and apical dominance; gibberellins: seed germination and stem elongation; cytokinins: cell division and side-bud growth; abscisic acid: dormancy and closing stomata; ethylene: fruit ripening). Oral probes: 'Which hormone did

	Kenyan example in a table. Examples on the cards: tea pruning and plucking in Kericho to make bushes branch, bean and maize seed germination boosted by gibberellins, rooting of cuttings with natural sources of hormones such as bulb onion, Aloe and cinnamon in a school tree nursery, and ethylene ripening bananas and mangoes bought at Mombasa markets, and abscisic acid keeping dormant seeds from germinating in the dry season. (3) Pruning demonstration: the teacher shows two bean seedlings set up two weeks earlier, one with its tip pinched off and one intact, and learners measure height and count side branches, recording in a table.	say: 'Read together. Underline the function of your hormone and one Kenyan example.' Clarify difficult terms on the board: 'hormone = chemical messenger', 'apical = at the top', 'dominance = control'. For the demonstration say: 'Compare the two plants. Count the side branches and measure the height with a ruler.' Safety: only the teacher handles the blade. Ask: 'Which plant is taller? Which is bushier? What might explain it?' Cold-call 4 learners for measurements, and record on the board.		you read about and what does it do?' Misconception to catch: 'Hormones are only found in animals.'
Explain Phase  VIDEO: Growth models: introduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners build a Claim-Evidence-Reasoning (CER) statement in their groups to answer: 'How do hormones help explain why maize and bean plants grown together differ in germination, emergence and growth?' They write the claim (e.g. 'Hormones such as gibberellins, abscisic acid and auxins control when seeds germinate and how the shoot and branches grow, so differences in hormone balance and response can cause different germination and growth'), then list at least two pieces of evidence (the pruned versus intact bean seedling data, the video or chart observations, the reading on abscisic acid and gibberellins in dormancy, and earlier lessons on dormancy and germination) and finally write the reasoning linking evidence to claim (e.g. 'When the tip, the source of auxin, was removed, side buds were released, showing that auxin	Say: 'A good scientific explanation has three parts: a claim, evidence and reasoning. Let us build one together.' Model one example on the board with a different question (e.g. 'Why do tea bushes in Kericho become bushy after pruning?'). Then say: 'Now do it for the maize and bean question.' Circulate and ask: 'What is your evidence for that?' and 'How does that evidence support your claim?' WAIT TIME 10-15 seconds after each question before offering any help. Cold-call 3 groups to read their claims and 3 different learners to give a piece of evidence. Ask the class: 'Do hormones act alone, or do they work together? Think about abscisic acid and gibberellins in seeds.' Take 2 more responses. Probe misconceptions: 'Does cutting the tip kill the plant? What did our data show?' Provide an extension challenge to faster	Claim-Evidence-Reasoning (CER) with peer review: learners must justify claims with data they collected and explicitly connect the data to a scientific idea (apical dominance and hormonal control). Modelling the structure first reduces cognitive load and supports learners writing in a second language, while peer review teaches them to critique evidence, not opinions. Connecting to agriculture shows that the explanation has real-world value in Kenyan farming and tree nurseries.	Peer review checklist and teacher scoring rubric (0-3) for each CER: claim present and answers the question; at least two pieces of evidence taken from data or reading; reasoning links evidence to hormone action. Success indicators: learners use terms such as 'auxin', 'apical dominance', 'gibberellin' and 'abscisic acid' correctly; evidence cites actual numbers (e.g. '4 side branches versus 0'). Oral probing identifies misconceptions: (a) 'hormones are the only factor in germination', (b) 'removing the tip kills the plant', (c) 'one hormone does one job only'. Quick quiz (3 questions, 3 minutes): name the hormone that breaks dormancy; state what apical dominance means; give one agricultural use of ethylene or pruning. Aim for 70% correct.

	from the tip suppresses side branches; this is apical dominance. Because maize is a monocot with one main stem and beans are dicots that can branch, their hormone responses and growth forms differ.'). Groups also explain farm applications: why tea farmers in Kericho prune bushes, why a nursery manager may dip cuttings in onion or Aloe extract, and why Mombasa traders put bananas in a closed sack to ripen them faster. Each group swaps its CER with another group for peer review using a three-point checklist: Is the claim clear? Is the evidence from our data? Does the reasoning link both?	groups: 'How could a farmer use this knowledge to get more branches on bean plants or more roots on a cutting?' Close by saying: 'Hormones are one part of the answer; the other parts are the seed structure, dormancy, water, oxygen and temperature from our earlier lessons.'		
Driving Question Board (DQB) Creation  VIDEO: Phototropism Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board with the main question: 'What causes the differences in germination, emergence and growth of maize and bean plants?' Each group writes (a) one new answer or piece of evidence from today on a green sticky note or paper strip, e.g. 'Gibberellins and abscisic acid control dormancy and germination' or 'Auxin from the tip causes apical dominance', and (b) one new question on a different colour, e.g. 'How do hormones interact with water and temperature?' or 'Can we use onion or Aloe to root cuttings for our school nursery?'. Learners match their notes to earlier questions on the board and move any questions that have now been answered into the 'Answered' column. Learners also complete an exit ticket: 'What evidence did we gather today?' and 'How does this help answer our driving question?'	Say: 'Let us see how today's evidence helps our big question. Each group, one answer and one new question.' Allow 3 minutes for writing. WAIT TIME 10-15 seconds after asking: 'Which of our earlier questions can we now move to Answered?' Cold-call 3 groups to place a note and explain the link in one sentence. Ask: 'Which questions are still not answered?' and highlight 2 for Lesson 10 (e.g. 'How do all the factors work together?'). Link to the project: 'Our nursery project will use natural hormone sources such as onion, Aloe and cinnamon on stem cuttings. What would you test?' Collect exit tickets at the door.	Discussion and questioning with the Driving Question Board: sorting and connecting questions shows learners the unit as a coherent story. Writing new questions encourages curiosity and gives the teacher information to shape Lesson 10. The exit ticket provides individual accountability for explaining how evidence connects to the driving question.	Exit ticket scoring: 2 points if the learner names specific evidence (e.g. 'the bean with its tip removed had more branches') and links it to the driving question; 1 point if only one part; 0 if blank or vague. Look for new questions that show deeper thinking (interaction of factors, applications). Observation: note groups where every member contributes to the notes. Oral probe: 'Which earlier lesson connects to today's hormone idea?' Use results to identify learners needing a short re-teach at the start of Lesson 10.

Model Building Phase  VIDEO: Phototropism Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Hormones and Plant Growth - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners take out their Day 1 drawings and explanations of why the maize and bean seeds behave differently. Individually, they revise the model by adding: (a) hormone labels and arrows (gibberellins promoting germination and stem elongation, abscisic acid maintaining dormancy, auxin from the shoot tip suppressing side buds, cytokinins promoting side bud growth and cell division); (b) a note on how hormones interact with water, oxygen, temperature, seed structure and dormancy from previous lessons; and (c) a short sentence beginning 'I changed my model because...'. They then take part in a Gallery Walk: groups display their revised models with seed structure drawings, germination results, growth curves and CER explanations. Learners walk around in rotation with sticky notes, leaving one 'star' (a strength) and one 'wish' (a suggestion) at each display. Finally, learners return to their own work and make one further improvement based on peer feedback.	Say: 'Look at your Day 1 model. What was missing? Add the hormones we learned about today and show how they connect to the other factors.' Give 10 minutes of individual work. Circulate and ask: 'Where on your model does the signal start? What does it do?' WAIT TIME 10-15 seconds before responding to learners who hesitate. Then organise the Gallery Walk: 'Move clockwise every 2 minutes. Leave one star and one wish.' Cold-call 4 learners afterwards: 'What feedback helped you improve your model?' and 'What change did you make because of it?' Highlight strong models, particularly those linking hormones to both germination and growth, and say: 'Next lesson we will bring everything together into one final explanation and test our claim against all the evidence.' Remind learners to bring seedlings or notes from the nursery project.	Model revision with Gallery Walk and peer review: revisiting Day 1 models and explicitly stating what changed helps learners see their own learning (metacognition, Learning to Learn). Peer feedback through stars and wishes builds Unity and develops the habit of critiquing ideas using evidence. Integrating hormones with previously studied factors guides learners toward a systems view of plant growth.	Model check indicators: the revised model includes at least three correctly labelled hormones with functions; arrows show the direction of cause and effect (e.g. tip, then auxin, then side buds suppressed); at least two earlier factors (water, oxygen, temperature, dormancy, seed structure) appear connected to hormones; the learner writes a clear 'I changed my model because...' statement. Peer review: star and wish notes are specific (e.g. 'Your arrow from the tip is clear' or 'Add where gibberellins act'). Teacher uses the observation checklist to record participation and feedback quality. Misconception check: models that show hormones acting alone, with no role for water or temperature, are flagged for discussion at the start of Lesson 10.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners connect hormones to the phenomenon of different germination and growth in maize and beans, or did they treat hormones as an isolated topic? (2) Which hormones were hardest for learners, and what examples (tea in Kericho, bananas in Mombasa, cuttings in a school nursery) helped most? (3) Were the pruning data clear enough to support CER claims, and would a longer observation time or a second demonstration have improved it? (4) Did cold-calling and wait time bring quieter learners into the discussion? (5) Were group sizes and shared reading cards manageable, and how did the offline alternatives work with no electricity? (6) Which misconceptions remain (hormones act alone; cutting the tip kills the plant; hormones exist only in animals), and how will they be addressed in Lesson 10? (7) Which questions on the Driving Question Board should shape the final synthesis? Record learner questions about the tree nursery project for follow-up.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a bean seedling with its tip removed grew more side branches but was shorter, while the intact seedling grew
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	taller with few branches. We also saw (in the video or chart) how plant parts respond to hormones, e.g. bending toward light and rooting of cuttings, and we read that gibberellins, abscisic acid, cytokinins, auxins and ethylene each control different parts of growth.
What did I learn?	We learned that hormones are chemical messengers that control germination, dormancy, cell elongation, branching, rooting and fruit ripening. Auxin from the shoot tip causes apical dominance, and removing the tip releases side buds. Farmers use this knowledge when pruning tea in Kericho, rooting cuttings in tree nurseries and ripening bananas and mangoes.
How does this explain the phenomenon?	Hormones help explain why maize and beans planted together on the same day can germinate, emerge and grow differently: the balance of gibberellins and abscisic acid affects whether a seed breaks dormancy, and auxin and cytokinins control shoot growth and branching, with differences in species and seed condition changing the outcome. These internal signals work together with water, oxygen, temperature and seed structure to produce the differences we saw in the farmer's field.

LESSON 10 (80 minutes (double period)): Synthesis and Application: Explaining the Farmer's Maize and Beans and Planning the Tree Nursery

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners synthesise evidence from Lessons 1-9 to write a final Claim-Evidence-Reasoning explanation of why maize and bean seeds planted on the same day after rain differ in germination, emergence and growth. They complete and compare their final model with the Day 1 model, close out the Driving Question Board, and apply their learning to plan a tree nursery using stem cuttings and natural rooting hormones (bulb onion, Aloe, cinnamon).
Knowledge	By the end of the lesson, the learner should be able to: (a) explain how seed structure (monocot maize vs dicot bean), dormancy, germination conditions (water, oxygen, suitable temperature), type of germination (hypogeal in maize, epigeal in common bean) and hormones (auxins, gibberellins, cytokinins, abscisic acid) together account for differences in germination, emergence and growth; (b) distinguish growth from development and describe primary and secondary growth and apical dominance; (c) apply knowledge of hormones to plan rooting of stem cuttings in a tree nursery.
Skills	Constructing explanations (NGSS SEP 6) using Claim-Evidence-Reasoning; developing and using models (SEP 2) by revising the Day 1 model; analysing and interpreting data (SEP 4) from germination tables and growth curves; engaging in argument from evidence (SEP 7) during the Gallery Walk; planning investigations (SEP 3) for the nursery project; communicating scientific information in simple, clear English.
Attitudes	Unity: learners work harmoniously in groups to build and review the final explanation. Patriotism: learners value sharing seedlings with the school and community for environmental conservation. Learning to Learn: learners reflect on how their thinking changed since Day 1 and adjust accordingly. Citizenship: learners plan to solve community problems by sharing nursery seedlings. Financial Literacy: learners explore generating income by selling seedlings at a subsidised price. Prevention of Global Warming: learners recognise the role of tree planting in mitigating climate change. Appreciation of the importance of growth and development in plants.
Key Inquiry Question	How does growth occur in plants? What causes the differences in germination, emergence and growth of maize and bean plants?
Purpose in Storyline	Lesson 10 is the culminating lesson. Learners have gathered evidence on seed structure, dormancy, germination conditions, types of germination, primary and secondary growth, growth measurement and hormones. Now they bring everything together to answer the Driving Question about the farmer's maize and beans, compare their Day 1 and final models, and transfer their understanding to a real-life project, the tree nursery, which links to the KICD project and to community service.
Safety Notes	Use only seeds, cuttings and materials that are untreated; seeds dressed with pesticide (often pink or purple coated) must not be handled or tasted. Wash hands after handling soil, onion juice, Aloe gel and cinnamon. Do not taste any materials. Use a clean, sharp knife or razor only under teacher supervision for cutting stems (teacher or one designated learner per group cuts at an angle; others observe). Keep Aloe sap away from eyes. Learners with allergies to onion, cinnamon or Aloe should observe and record rather than handle. Move carefully during the Gallery Walk to avoid collisions in crowded classrooms.

B. LESSON OVERVIEW

This final lesson asks learners to answer the question: why did the farmer's maize and beans, planted on the same day after the same rain, germinate, emerge and grow differently? Learners begin by predicting how a farmer in Kakamega or Nyeri could improve germination and rooting of new plants. They then review the evidence collected across the unit (class germination tables, seed drawings, growth curves, hormone notes) in an evidence-sorting activity using Think-Pair-Share. In the Explain

phase, they write a Claim-Evidence-Reasoning (CER) final explanation and take part in a Gallery Walk with peer review. The class then closes the Driving Question Board (DQB), sorting questions into answered, partly answered and new questions. Finally, learners compare their Day 1 model with the final cumulative model and plan the tree nursery project using stem cuttings with bulb onion, Aloe and cinnamon as natural rooting hormone sources. The lesson requires only chart paper, previous learner notebooks, class data, simple drawings and locally available materials; no electricity or internet is required, and time-lapse video can be replaced with the teacher's printed photo sequence.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Linear vs. exponential growth: from data Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the anchoring phenomenon: the farmer in Bungoma plants maize and beans on the same day after rain, but some seeds germinate quickly, some slowly, and some not at all even though they look healthy. Using the teacher's printed photo sequence (or the time-lapse video if available), they view the growth sequence once more. Individually they write a 3-minute prediction on a half sheet: 'If the farmer wanted to grow 200 mango and grevillea seedlings from stem cuttings for a school tree nursery in Machakos, what would help the cuttings form roots, and why?' Learners predict using what they know about auxins and plant growth. They then Think-Pair-Share with a partner and add one idea from the partner in a different colour.	Display the Day 1 model charts and the DQB at the front. Say: 'Today is our final lesson. Today we will answer the farmer's puzzle and use our learning to start a tree nursery.' Show the photo sequence for 2 minutes. Ask: 'What puzzled us on Day 1?' Give WAIT TIME of 10-15 seconds. Cold-call 3 learners to recall the puzzle. Pose the nursery prediction question and give 3 minutes of silent writing time. Say: 'Think first. Write your own idea before you talk.' Then give 2 minutes for pair talk. Circulate and listen for mention of hormones, auxins, cut surface, water, or light. Cold-call 4 pairs to share. Record predictions on the board under 'Our Nursery Predictions'. Do not correct yet; say: 'We will test these ideas against our evidence today.'	Predict-Observe-Explain (POE) and Think-Pair-Share: the prediction step surfaces learners' current ideas about hormones and rooting before they apply evidence, so that transfer of learning to a new context (stem cuttings) can be checked. Think-Pair-Share gives every learner thinking time and a low-risk chance to rehearse ideas, which supports participation in large classes.	Observation checklist: tick learners who (1) write a prediction with a reason, (2) mention at least one hormone or growth factor such as auxins, water or oxygen, (3) add a partner's idea in a different colour. Oral probing: ask 'Why do you think that would help roots form?' Look for misconceptions such as 'cuttings need fertiliser to root' or 'hormones are only found in animals'. Record names of 5 learners whose predictions need follow-up.
Observe Phase  VIDEO: Linear vs. exponential growth: from data Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML:	In groups of 5 or 6, learners review the evidence collected across Lessons 1-9 using an Evidence Sorting Station. Each group receives a set of cards or uses their notebooks showing: (1) labelled drawings of maize (monocot) and bean (dicot) seeds, (2) class germination data for maize and beans under different conditions (water, no water, with and without air, cold vs warm), (3)	Arrange evidence cards or ask learners to open notebooks to the marked pages. Say: 'You are scientists. Your job is to find which evidence explains the farmer's puzzle.' Give each group 10 minutes. Circulate and ask: 'Which piece of evidence shows why some seeds did not germinate?' and 'Where do you see the effect of water or oxygen in our data?' Use WAIT TIME of 10-15 seconds	Evidence sorting with Discussion and questioning: categorising prior evidence into seed, environment and hormone factors helps learners see that many causes act together rather than one cause alone, and prepares them to build a multi-part explanation. Observing the rooting results gives a new data set that tests whether their understanding of hormones transfers to a new context.	Observation checklist indicators: (1) group places at least 6 of 8 evidence items under the correct heading, (2) learners point to specific data (e.g. 'Beans in dry cotton wool did not swell') when justifying, (3) learners read the rooting table accurately and identify the control. Oral probing: 'Why do we need a cutting with only water?' Listen for the idea of a control. Quick check: ask each

Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	results on dormancy-breaking (soaking, scarification), (4) hypogeal vs epigeal germination observations, (5) a seedling height growth curve, (6) notes on primary and secondary growth and (7) notes on auxins, gibberellins, cytokinins, abscisic acid and apical dominance. Groups sort evidence under three headings: 'Seed factors', 'Environmental factors' and 'Hormone and growth factors'. They then examine a set of prepared stem cuttings or photographs: hibiscus or sweet potato vine cuttings treated with crushed bulb onion, Aloe gel, cinnamon powder or water only (control), prepared by the teacher or earlier by the class, and record number of roots and days to rooting in a simple table.	after each question. Cold-call 2 members from different groups per question. When groups examine the cuttings, ask: 'What is different about the treated and untreated cuttings? What could be the cause?' If the real cuttings are unavailable, use the teacher-prepared table of results from the earlier class experiment. Remind learners to use simple scientific English, e.g. 'The bean seed absorbed water, so the seed coat softened.' Clarify difficult terms such as 'scarification', 'epigeal' and 'apical dominance' on the board. Move quickly between groups to ensure equipment and notebooks are shared fairly.		group to name one piece of evidence for each of the three headings. Note groups that confuse dormancy with lack of water.
Explain Phase  VIDEO: Linear vs. exponential growth: from data Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	Learners write a Final Explanation using the Claim-Evidence-Reasoning (CER) frame, individually after group discussion, to answer the Driving Question: 'What causes the differences in germination, emergence and growth of maize and bean plants?' CLAIM: a clear statement naming the causes (e.g. seed structure and food stores, dormancy status, availability of water, oxygen and warmth, type of germination, hormone levels). EVIDENCE: at least three specific pieces of data from the unit (e.g. 'In our class test, 8 out of 10 soaked beans germinated compared with 3 out of 10 dry beans'; 'Maize remained underground with cotyledon (hypogeal) while bean cotyledons rose above ground (epigeal)'). REASONING: link evidence to biology ideas (e.g. 'Water activates enzymes and gibberellins that	Display the CER sentence frames on the board: 'We claim that ... Our evidence is ... This happens because ...'. Say: 'A good scientific explanation uses evidence from our own investigations, not just what we remember.' Give 12 minutes of writing time, with the first 2 minutes in silence. Circulate and ask: 'Which data from our class results supports that claim?' and 'Explain the link: why does that evidence make your claim true?' Use WAIT TIME of 10-15 seconds. Before the Gallery Walk, model peer feedback with one example: 'Star: you used our germination numbers. Step: explain why water is needed.' Set the Gallery Walk for about 12 minutes, with groups rotating on a clap signal. Cold-call 4 learners afterwards to share one improvement they will make. Listen for misconceptions such as	Claim-Evidence-Reasoning (CER) builds understanding by requiring learners to connect data to scientific principles rather than list facts. The Gallery Walk with peer review engages learners in argument from evidence: they compare explanations, notice missing links and revise, which deepens understanding of multiple interacting causes. Revising after feedback supports Learning to Learn.	Peer review rubric (3 levels) for CER: Claim names at least 3 causes; Evidence cites at least 3 specific class data or observations; Reasoning links each evidence item to a biological principle (water uptake, enzyme activation, hormone action, apical dominance). Observable indicators: learners use words such as 'because' and 'therefore', cite numbers from tables, and revise after feedback. Oral probing during revision: 'What did you change after the Gallery Walk?' Collect 5 CER explanations at random as a quick quiz-style check and note common gaps for later feedback.

	break down stored food, so the embryo can grow; the epigeal bean raises its seed leaves, so it emerges differently from maize'). Learners also explain differences in later growth: maize shows mostly primary growth and tall unbranched stem, while bean shows branching affected by apical dominance and auxins. After writing, groups pin their CER explanations and drawings on the wall for a Gallery Walk. Learners rotate every 3 minutes with sticky notes giving one 'star' (strength) and one 'step' (improvement) per display. They then return to their own work and revise it using the feedback.	'Maize has no dormancy' or 'hormones make seeds grow without water'. Address them: 'Check your evidence. Can a seed germinate without water in our test?' Allow learners to revise for 5 minutes. Use a whole-class check with a Kenyan example: 'A farmer in Kitale plants hybrid maize seeds that failed to germinate after planting in dry soil. What advice would you give?' Cold-call 3 learners.		
Driving Question Board (DQB) Creation  VIDEO: Linear vs. exponential growth: from data Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board created in Lesson 1, which holds their questions (for example, 'Why did some healthy-looking seeds fail to germinate?', 'Why do bean cotyledons come above the ground but maize ones do not?', 'What makes stems grow taller and thicker?', 'How do hormones affect rooting?'). In groups, they take turns choosing a question card, deciding whether it is Answered, Partly answered or New, and placing it in the corresponding column on the board. For each answered question, a learner writes a one-sentence answer with the evidence and lesson number on a sticky note (e.g. 'Lesson 4: Beans in water at room temperature germinated; those with no water did not'). Learners also add new questions arising from the nursery project, such as 'Which natural hormone gives the most roots on mango cuttings?' or 'Can we sell seedlings at a profit if we use free	Point to the DQB and say: 'Let us check how far our questions have been answered.' Invite 6 groups, each to handle 2-3 questions. Say: 'Choose a question. Tell us what evidence from which lesson answers it.' Give WAIT TIME of 10-15 seconds before calling on a group to defend its placement. Cold-call 5 learners to justify a card placement. If a question is partly answered, ask: 'What else do we need to find out?' Encourage learners to link to the farmer's puzzle: 'How does this answer help explain what happened in the farmer's field in Bungoma?' Give 3 minutes for adding new questions. Hand out exit tickets and give 4 minutes to write. Collect them at the door. Review a few tickets immediately to identify misconceptions to address next lesson.	Discussion and questioning with the Driving Question Board: sorting questions by how well they have been answered makes learning visible and helps learners see how each lesson contributed to the main question. Adding new questions models how science generates further inquiry, and motivates the project phase.	Observable indicators: learners correctly place at least 80 percent of cards, cite the specific lesson and data to justify, and suggest at least one new relevant question. Exit ticket success criteria: names at least two pieces of evidence (e.g. germination counts, growth curve, rooting results) and links them to the driving question. Oral probing: 'Which question is still partly answered and why?' Note learners whose tickets list facts without linking them to the driving question.

	onion and Aloe?'. They then complete a short exit ticket: 'What evidence did we gather today?' and 'How does this help answer our driving question?'			
Model Building Phase  VIDEO: Linear vs. exponential growth: from data Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Seed Dormancy and Germination - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Learners take out their Day 1 model (a drawing and short explanation of why the farmer's maize and beans behaved differently) and place it next to a new, large final cumulative model drawn in their groups on chart paper. The final model shows: (1) labelled maize and bean seeds (seed coat, embryo, cotyledon(s), endosperm); (2) factors triggering germination (water, oxygen, warmth, dormancy breaking), with arrows to hormones (gibberellins, abscisic acid) in the seed; (3) hypogeal germination in maize and epigeal germination in bean with emergence heights; (4) primary growth at root and shoot tips, secondary growth in woody dicots, and apical dominance with auxins; (5) a growth curve (S-shaped) for seedling height. Using a different colour pen, learners mark on the Day 1 model what was missing or incorrect and write 'Then I thought... Now I know... because...' statements (at least 3). Finally, groups apply the model to the tree nursery project by drawing a plan for rooting stem cuttings of mango, grevillea or hibiscus: selecting healthy cuttings 15-20 cm, cutting just below a node, applying crushed bulb onion juice, Aloe gel or cinnamon powder (natural rooting hormone sources), planting in moist sandy soil in polythene tubes, shading, watering daily, and recording rooting. They list roles (Unity), a basic budget and a subsidised selling price (Financial	Say: 'Hold up your Day 1 model. What would you change now?' Allow 15 seconds of WAIT TIME. Cold-call 3 learners to name one change. Give groups 12 minutes to draw the final model and 5 minutes for the 'Then I thought... Now I know...' statements. Circulate and ask: 'Where does the hormone appear in your model, and what does it do?' and 'How does your model explain why maize cotyledon stays below ground?' Push for links: 'Where is the evidence for that arrow?' Then introduce the nursery plan: 'Our model says auxins stimulate root formation. How can we use onion, Aloe and cinnamon to help our cuttings root?' Give 8 minutes for planning, and remind learners about safety (no tasting, wash hands, teacher or one trained learner cuts stems). Cold-call 4 groups to present one key step and its scientific reason. Close by saying: 'You can now explain the farmer's puzzle and help your community plant trees.' Assign the nursery set-up as a follow-up task: groups bring cuttings, polythene tubes or tins and soil for the next practical session, and record weekly rooting data. Collect group models for a quick model check.	Model revision: comparing the Day 1 model with the final model makes conceptual change visible, so learners can see how evidence added mechanisms (hormones, dormancy, germination type, growth regions) to their initial ideas. Applying the model to the tree nursery is a transfer task that shows whether learners can use scientific explanations to solve a real community problem.	Model check indicators: final model includes at least 5 of the 6 required components (seed structures, germination conditions, hormones, hypogeal vs epigeal, primary/secondary growth and apical dominance, growth curve), labels are accurate, arrows show cause and effect, and at least 3 evidence-based 'Then I thought... Now I know...' statements are present. Nursery plan checklist: includes a control (water only), a hormone source with scientific reason, safe handling, and roles. Peer review: groups swap plans and give one star and one step. Record learners who still hold misconceptions (for example, 'hormones feed the cutting') and plan follow-up feedback.

	Literacy), and where seedlings will be planted or shared in the community (Citizenship, Patriotism, Prevention of Global Warming).			
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D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Did learners' final CER explanations include multiple interacting causes (seed structure, dormancy, water, oxygen, warmth, germination type, hormones) rather than a single cause? Which causes were most often missing? (2) How did learners' final models differ from their Day 1 models, and which learners showed the greatest conceptual change? (3) Which misconceptions persisted, such as confusing growth with development, thinking hormones are food, or thinking all seeds germinate in the same way? (4) Did the Gallery Walk and peer review help learners improve their explanations, and was the time sufficient in a large class? (5) How well did learners transfer their understanding of auxins to the tree nursery plan, and what support will they need to set up and monitor the nursery (materials, supervision, record keeping, and selling seedlings at a subsidised price)? (6) Did all learners, including those with limited access to textbooks, participate fully? Plan how to follow up weekly with the nursery data and revisit the DQB for remaining questions.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that maize and bean seeds differ in structure (monocot with endosperm vs dicot with cotyledons), that seeds failed to germinate without water, oxygen or warmth or when dormant, that maize germinates hypogeally and bean epigeally, that seedling height follows an S-shaped growth curve, and that stem cuttings treated with natural hormone sources such as bulb onion, Aloe or cinnamon rooted differently from the control.
What did I learn?	We learned that germination, emergence and growth depend on seed structure and food stores, dormancy and its breaking, water, oxygen and suitable temperature, type of germination, growth at meristems (primary and secondary growth), and plant hormones such as auxins, gibberellins, cytokinins and abscisic acid, including apical dominance. We also learned that growth is an increase in size while development involves changes in form and function.
How does this explain the phenomenon?	The farmer's maize and beans differed because each seed had a different structure, dormancy status and germination type, and experienced different water, oxygen and warmth conditions around the seed. These factors changed the levels and action of hormones such as gibberellins and abscisic acid in the seed and auxins in the growing plant, which control when the embryo grows, how the seedling emerges, and how the stem and roots grow. We can apply this to grow tree seedlings from cuttings by using natural rooting hormone sources, and share the seedlings to benefit our community and help reduce global warming.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.